

Learning heterogeneous network representations for relation prediction via characterizing hierarchical and anisotropic generation process

Xuan Guo ^{a,b,c}, Qiyao Peng ^{a,b,c}, Wenjun Wang ^{a,b,c}, Yaozhi Zhang ^a, Zihao Liang ^a, Wei Yu ^d, Ying Zhao ^e, Tianpeng Li ^{f,b,c,*}

^a School of Artificial Intelligence, Tianjin University, Tianjin, 300350, China

^b State Key Laboratory of Synthetic Biology, Tianjin University, Tianjin, 300072, China

^c NHC Key Laboratory of Cutting-edge Biotechnology Policy Research, Tianjin University, Tianjin, 300072, China

^d School of Computer Science and Technology, Tianjin University, Tianjin, 300350, China

^e ENN Group, Langfang, Hebei, 065001, China

^f School of New Media and Communication, Tianjin University, Tianjin, 300350, China

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ABSTRACT

Heterogeneous networks, characterized by diverse node and edge types, present unique challenges for representation learning. Existing methods predominantly focus on local pairwise relations such as meta-paths, a flat modeling paradigm that neglects the inherent hierarchical and higher-order structures governing network formation. Such limitations often result in suboptimal representations that fail to preserve the global topology. To address these issues, we propose REHHOR (Reconstructive Embedding via Hierarchical Higher-Order Relations), a novel framework designed to explicitly mirror the hierarchical generation process of heterogeneous networks. REHHOR focuses on the cohesive structural primitive, i.e., the Higher-Order Unit (HOU) and decomposes complex interactions into type-specific node-HOU bipartite graphs. To accommodate the anisotropic adjacency patterns observed in these structures, we introduce a directed diffusion mechanism that injects data-dependent noise to generate node and HOU representations. Critically, rather than maximizing traditional pairwise similarities, REHHOR optimizes embeddings by explicitly reconstructing the interaction skeleton formed by interconnected HOU. This synergistic design of hierarchical encoding and skeletal reconstruction enables the model to capture the overarching structural dependencies often lost in local modeling. Extensive experiments on diverse real-world datasets demonstrate that REHHOR consistently outperforms state-of-the-art baselines across multiple relation prediction tasks, particularly exhibiting superior robustness in scenarios with data sparsity.

1. Introduction

Real-world systems, ranging from academic collaborations to financial transactions, are naturally represented as heterogeneous networks in which diverse node and edge types encode multifaceted entity interactions (Yang et al., 2022). These edges constitute the fundamental building blocks of network topology and define the underlying structural dependencies. Consequently, identifying missing links or predicting potential relations has emerged as a primary challenge in network analysis (Xu, 2021). In heterogeneous environments, this task is further complicated by the necessity of integrating diverse semantic information across heterogeneous node and relation types. Nevertheless, such complexity provides a richer context for uncovering the intrinsic mechanisms of the system. Accurate relation prediction thus facilitates various critical

applications, such as friend recommendation (Wang et al., 2023a), drug-target identification (Li et al., 2022a), and fraud detection (Jiang et al., 2021a).

Network representation learning has emerged as the primary paradigm for relation prediction (Xu, 2021). Early approaches utilized lookup tables (Dong et al., 2017) or classic neural networks (Qu et al., 2018) to optimize co-occurrence probabilities (Tang et al., 2015a) or geometric scoring functions (Bordes et al., 2013; Sun et al., 2019). Despite the success of Heterogeneous Graph Neural Networks (HGNNs) (Wang et al., 2019), current optimization strategies remain largely confined to node-centric local patterns, relying on proximity reconstruction (Fu et al., 2020) or contrastive consistency (Yu et al., 2024). Theoretically, although self-supervised learning can adaptively capture rich semantics, its encoding mechanism remains fundamentally tethered to the inherent

* Corresponding author.

E-mail addresses: guoxuan@tju.edu.cn (X. Guo), qypeng@tju.edu.cn (Q. Peng), wjwang@tju.edu.cn (W. Wang), yaozhi@tju.edu.cn (Y. Zhang), liangzihao@tju.edu.cn (Z. Liang), weiyu@tju.edu.cn (W. Yu), zhaoying@enn.cn (Y. Zhao), ltptimeia@tju.edu.cn (T. Li).

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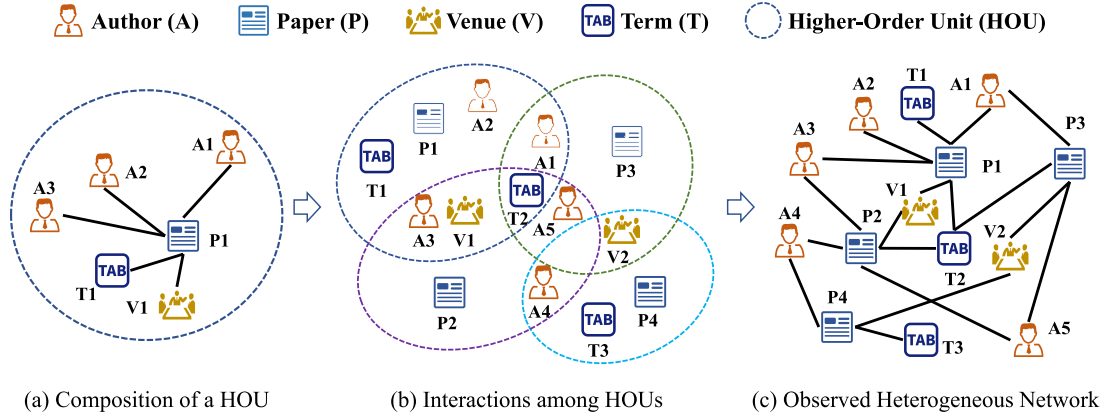


Fig. 1. The hierarchical generation process of heterogeneous networks. (a) Formation of a higher-order unit (e.g., a publication event) through fine-grained node interactions. (b) Interactions between higher-order units via shared nodes form the network skeleton. (c) The resulting global network topology.

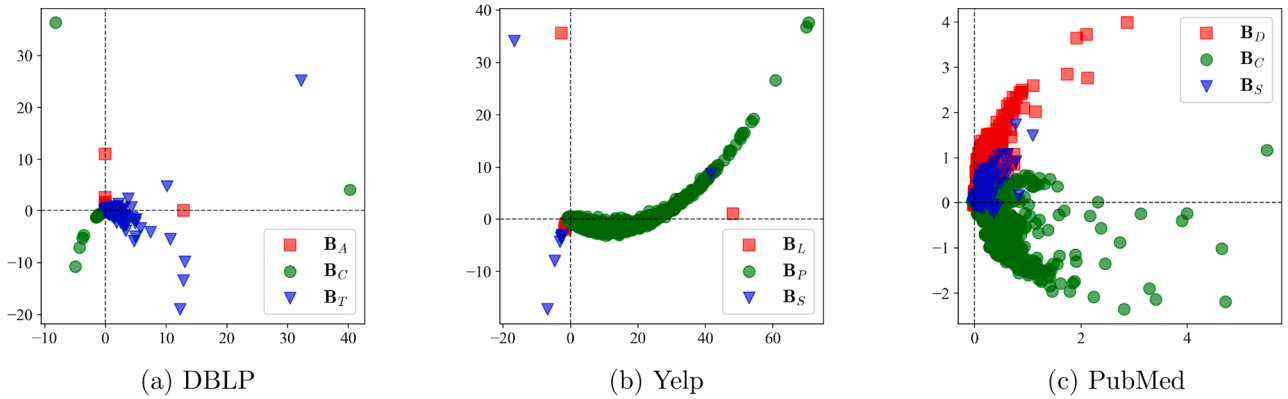


Fig. 2. PCA visualization of node-HOU adjacency relations. The distributions show elongated profiles rather than uniform spreading, signifying the strong anisotropy inherent in the heterogeneous topology of real-world networks.

tendency of HGNNs to prioritize pairwise relations. Consequently, these methods are primarily restricted to a flattened local perspective, an orientation that inherently fails to capture the essential structural dependencies in complex systems.

By fundamentally overlooking the hierarchical assembly process of real-world networks, these flat, pairwise modeling paradigms fail to effectively characterize the global topology. Specifically, fine-grained node interactions do not exist in isolation but first aggregate into higher-order functional units (HOUs). These HOUs subsequently intertwine to form the overarching network topology. As illustrated in Fig. 1, using an academic network as an example, individual publications emerge not as mere sets of independent pairwise links but as unified higher-order events arising from concurrent interactions among authors, keywords, and venues. These events (HOUs) serve as the fundamental semantic building blocks whose subsequent interplay through shared nodes shapes the global collaboration structure. Characterizing this hierarchical generation is essential for preserving the structural integrity of the network, which is particularly vital for achieving robust relation prediction under data-sparse conditions.

Despite recent progress in exploring higher-order structures (Fan et al., 2021; Khan et al., 2025), current methods face the following two fundamental limitations that prevent them from fully capturing the hierarchical generation process:

(1) Higher-Order Semantic Homogenization. While existing methods distinguish node types at the neighborhood level, they often overlook the heterogeneity inherent in the formation of higher-order structures. Most hypergraph-based models collapse heterogeneous nodes into a unified higher-order representation, erasing the functional

distinctions between different local relation types (Fan et al., 2021). Crucially, these hypergraph methods only reconstruct node-hyperedge associations, treating each hyperedge as an isolated set and ignoring the interactions between HOUs. This oversight prevents them from capturing the global skeleton formed by HOU-HOU connectivity.

(2) Anisotropic Modeling Inconsistency. As illustrated in Fig. 2, Principal Component Analysis (PCA) of the adjacency patterns in node-HOU bipartite graphs reveals that node associations are not uniformly distributed but instead exhibit a striking phenomenon of anisotropy. These associations concentrate along specific semantic directions that correspond to dominant relation types. However, conventional generative frameworks such as variational autoencoders, normalizing flows, and emerging diffusion models predominantly rely on isotropic noise assumptions or spherical Gaussian priors. This fundamental discrepancy between the anisotropic data nature and isotropic modeling assumptions induces significant signal degradation, thereby impeding the faithful recovery of type-specific connectivity patterns.

(3) Global Skeleton Negligence. Current optimization objectives primarily focus on preserving local node similarities or neighborhood structures. Even higher-order models tend to prioritize the reconstruction of local relations rather than the explicit modeling of the interaction skeleton formed by HOUs (Fan et al., 2021). This focus results in a significant loss of robustness when local connectivity is compromised. Specifically, the underlying global skeleton often remains resolute even when local relations are sparse or noisy, providing a robust foundation for structural inference.

To address these limitations, we propose REHHOR (Reconstructive Embedding via Hierarchical Higher-Order Relations), a framework

designed to faithfully model the hierarchical generation process of heterogeneous networks. Unlike hypergraph-based methods that treat HOU as independent sets, REHHOR explicitly models HOU-HOU interactions via shared non-center nodes and optimizes a skeletal contrastive objective. This shift from local to skeletal reconstruction is the core methodological advance. The framework consists of three synergistic components:

- **Type-Specific Bipartite Decomposition.** By constructing separate bipartite graphs for each non-center node type, REHHOR maintains the functional roles of different entity types within HOU, avoiding the homogenization problem of hypergraph models.
- **Hierarchical Skeletal Learning.** REHHOR first decomposes higher-order interactions into multiple type-specific node-HOU bipartite graphs to preserve semantic heterogeneity. It then employs a higher-order contrastive objective that operates directly on HOU-HOU interactions, forcing the model to preserve the global backbone formed by shared nodes. This contrasts with prior hypergraph methods that only reconstruct memberships between nodes and hyperedges, but lose inter-HOU connectivity.
- **Directed Diffusion for Anisotropic Adjacency.** Motivated by the PCA observations, we introduce a directed diffusion mechanism that injects data-dependent, directional noise into the node-HOU bipartite adjacency matrices. This preserves anisotropic structural signals and enables efficient single-step denoising. While directional noise has been explored for node features, its application to membership matrices for skeletal reconstruction is novel.

Extensive evaluations on real-world datasets demonstrate that REHHOR achieves state-of-the-art performance across multiple relation prediction tasks, exhibiting superior robustness particularly in the presence of data sparsity.

The main contributions of REHHOR are three-fold:

- We propose a comprehensive approach REHHOR to capture the hierarchical generation of heterogeneous networks. It reconstructs the interaction skeleton of higher-order units, rather than node-hyperedge associations or pairwise links. This is achieved through the harmonization of type-specific bipartite decomposition and skeletal-level contrastive optimization.
- We introduce a directed diffusion mechanism tailored to node-HOU bipartite adjacency matrices, incorporating data-dependent anisotropic noise to preserve structural signals—a first for membership matrix reconstruction in heterogeneous networks.
- We conduct comprehensive experiments, including a multi-center Hettionet subset, and provide ablation evidence that the hierarchical HOU structure and skeletal objective are individually necessary, addressing the reviewer concern that improvements might arise solely from increased model capacity.

2. Related work

This section reviews existing works on heterogeneous network representation learning and graph diffusion models, highlighting the limitations that motivate REHHOR.

2.1. Heterogeneous network representation learning

The inception of Heterogeneous Network representation learning was characterized by preserving local proximity via random walks or matrix factorization, as exemplified by Metapath2vec (Dong et al., 2017) and PTE (Tang et al., 2015a). This paradigm has been subsequently transcended by Heterogeneous Graph Neural Networks (HGNNs), fueled by the ascendancy of message-passing architectures (Bing et al., 2023; Yang et al., 2022). To capture intricate structural semantics, pioneering works such as R-GCN (Schlichtkrull et al., 2018), HAN (Wang et al.,

2019), and MAGNN (Fu et al., 2020) introduce relation-specific transformations and meta-path-based message passing. Attention-driven models like HGT (Hu et al., 2020), SR-HGN (Wang et al., 2023b), and HMAM (Liu et al., 2025) further refine node interactions by adaptively weighting hierarchical dependencies and cross-type interactions. Concurrently, to mitigate the burgeoning complexity of these models, streamlined architectures such as Simple-HGN (Lv et al., 2021) and SeHGNN (Yang et al., 2023b) have emerged to reconcile representational power with computational efficiency. To capture higher-order interactions, some methods like HeteHG-VAE (Fan et al., 2021) use hypergraphs to model group-level relations.

From the perspective of optimization, the field has evolved from local proximity reconstruction toward structured self-supervised learning (SSL). While early methods such as TransE (Bordes et al., 2013) and ConvE (Dettmers et al., 2018) focus on triplet-level alignment, contemporary SSL strategies bifurcate into contrastive and generative streams. The former, including CPT-HG (Jiang et al., 2021b), NHGMI (Li et al., 2024), MEOW (Yu et al., 2024), and HGCL (Yu et al., 2024), maximize mutual information across augmented views to learn structural invariance. The latter, exemplified by HGMAE (Tian et al., 2023) and HeteHG-VAE (Fan et al., 2021), focus on recovering latent topological or feature-level semantics through masked auto-encoding and variational inference.

Regarding relation prediction in heterogeneous networks, representation-based methods currently rely on measuring the affinity between node embeddings within a latent space optimized by the aforementioned architectures and objectives. However, these methods predominantly focus on modeling flat interaction patterns. They often neglect the hierarchical generation process where fine-grained node interactions aggregate into higher-order functional units before shaping the global network. This fundamental gap motivates the hierarchical structural modeling in REHHOR.

2.2. Graph diffusion models

Diffusion models have revolutionized generative modeling by simulating the stochastic transition between noise and structured data (Ho et al., 2020; Nichol & Dhariwal, 2021). On homogeneous networks, GRAND (Chamberlain et al., 2021) frames GNN layers as diffusion processes. DDM (Yang et al., 2023a) introduces specialized kernels to guide denoising flow along edge directions. DiGress (Vignac et al., 2023) enhances the expressiveness and sampling efficiency of graph generative processes via discrete denoising processes.

On heterogeneous networks, a pioneering effort is DiffGraph (Yang et al., 2025), which employs cross-view denoising to align auxiliary heterogeneous data with target semantic spaces. Despite such progress, diffusion-based modeling for heterogeneous networks is still in its infancy. Beyond DiffGraph and isolated applications in specialized medical domains (Wang et al., 2025a,b), generalized diffusion frameworks for complex multi-typed connectivity remain scarce. This leaves a significant research gap in leveraging diffusion processes to capture the intricate semantics of heterogeneous structures.

In this paper, our REHHOR framework introduces a directed diffusion mechanism specifically designed for the anisotropic generation process of heterogeneous networks. By explicitly characterizing the directional and type-specific nature of node-to-HOU associations in directed noise, REHHOR effectively preserves the skewed adjacency patterns inherent in heterogeneous structures, thereby providing a more robust generative foundation for high-fidelity relation prediction.

3. Preliminaries

This section establishes the formal foundations commonly used in this paper, with key notations and their descriptions summarized in Table 1.

Table 1
Main notations and their explanation.

Notation	Explanation
$H = (\mathcal{V}, \mathcal{E}, \Phi, \Psi, \mathcal{T}, \mathcal{T}_\mathcal{E})$	A Heterogeneous Network
$\mathcal{V}, \mathcal{E}, \mathcal{T}, \mathcal{T}_\mathcal{E}$	Sets of nodes, edges, node types and edge types
Φ, Ψ	Mapping from $\mathcal{V}$ to $\mathcal{T}$, mapping from $\mathcal{E}$ to $\mathcal{T}_\mathcal{E}$
$\mathcal{N}(v)$	Set of v 's neighbors
$S(H)$	Schema Graph corresponding to H
$\mathbf{z}_i \in \mathbb{R}^d$	d -dimension representation of node $v_i \in \mathcal{V}$
$o_c, \mathcal{T}_c$	Central node type, set of central node types
$\xi_v = (\mathcal{V}_v^\xi, \mathcal{E}_v^\xi), \Xi$	Higher-order unit (HOU), set of HOU's
$B_o = (\mathcal{V}_o^B, \mathcal{E}_o^B)$	Node-HOU Bipartite graph for type o
$\mathbf{B}^o$	Adjacency matrix of bipartite graph B_o
$\mathbf{h}_{i,o}$	Representation of higher-order unit ξ_{v_i} encoded on B_o
$\Xi_{i,\text{pos}}^o, \Xi_{i,\text{neg}}^o$	Set of positive/negative samples for contrastive loss
τ	Temperature parameter for contrastive loss

3.1. Heterogeneous network fundamentals

Definition 3.1 (Heterogeneous Network.). A heterogeneous network is defined as $H = (\mathcal{V}, \mathcal{E}, \Phi, \Psi, \mathcal{T}, \mathcal{T}_\mathcal{E})$, where $\mathcal{V}$ and $\mathcal{E}$ denote the sets of nodes and edges, respectively. The mapping functions $\Phi : \mathcal{V} \rightarrow \mathcal{T}$ and $\Psi : \mathcal{E} \rightarrow \mathcal{T}_\mathcal{E}$ associate each node and edge with their respective types, such that $|\mathcal{T}| + |\mathcal{T}_\mathcal{E}| > 2$.

In most heterogeneous networks, the type of an edge is determined by the types of its two endpoint nodes. Therefore, the discussion of edge types is sometimes omitted in this paper.

Definition 3.2 (Schema Graph.). Given a heterogeneous network H , its schema graph $S(H)$ summarizes all node type and relation type information, defined as $S(H) = (\mathcal{V}_\Phi, \mathcal{E}_\Psi)$, where $\mathcal{V}_\Phi = \{\Phi(v) \mid v \in \mathcal{V}\} \subseteq \mathcal{T}$ and $\mathcal{E}_\Psi = \{\Psi(e) \mid e \in \mathcal{E}\} \subseteq \mathcal{T}_\mathcal{E}$.

The schema graph $S(H) = (\mathcal{V}_\Phi, \mathcal{E}_\Psi)$ is the structural template of H , providing a high-level abstraction of the permissible node and relation types.

Definition 3.3 (Heterogeneous Network Representation Learning.). Given a heterogeneous network H , the objective is to learn a mapping function that projects each node $v_i \in \mathcal{V}$ into a low-dimensional vector $\mathbf{z}_i \in \mathbb{R}^d$ ($d \ll |\mathcal{V}|$). The obtained representation vectors $\{\mathbf{z}_i\}_{v_i \in \mathcal{V}}$ should capture as much informative knowledge as possible to facilitate downstream tasks.

3.2. Higher-order units

Real-world heterogeneous networks are typically composed of recurrent functional patterns. Within these patterns, certain node types act as hubs that orchestrate interactions among multiple entities. In such event-driven processes, these hub entities serve as unique identifiers or anchors of higher-order functional units. For example, in an academic network, a publication anchors the simultaneous presence of authors, venues, and keywords; in an e-commerce network, a business anchors reviews, star ratings, and locations. Such anchor entities correspond to node types that are directly connected to all other node types in the schema graph. Formally, we define them as central node types:

Definition 3.4 (Central Node.). Given a heterogeneous network H , if there exists a minimal set of node types $\mathcal{T}_c = \{o \mid o \in \mathcal{T}\}$ such that the node types in this set, together with all their neighboring node types in the schema graph $S(H)$, constitute the entire set of node types $\mathcal{T}$ in H , then the node types in $\mathcal{T}_c$ are called center node types.

In practice, we select central node types by first extracting the schema graph $S(H)$ from the network's metadata. Then, for each node type $t \in \mathcal{V}_\Phi$, we compute its neighbor set $\mathcal{N}_{S(H)}(t)$ in $S(H)$. If $\{t\} \cup \mathcal{N}_{S(H)}(t) = \mathcal{V}_\Phi$, then t is a central node type.

It is important to note that this definition is not limited to star networks. In heterogeneous networks where no single node type connects to all others, the center role may be fulfilled by a minimal set of node types.

Formally, a set $\mathcal{T}_c \subseteq \mathcal{T}$ is a **center type set** if $\mathcal{T}_c \cup \bigcup_{t \in \mathcal{T}_c} \mathcal{N}_{S(H)}(t) = \mathcal{T}$, and no proper subset of $\mathcal{T}_c$ satisfies this condition. The types in $\mathcal{T}_c$ form a connected subgraph in the schema graph $S(H)$. For example, in the biological information network that satisfies the schema graph shown in Fig. 3(b), both gene and compound nodes form the center node set of the network (Wang et al., 2021).

In a star network, each center node uniquely identifies a higher-order unit, which can be formally defined as follows.

Definition 3.5 (Higher-Order Unit (HOU).). For a central node v with $\Phi(v) = o_c$, its corresponding HOU is a subgraph $\xi_v = (\mathcal{V}_v^\xi, \mathcal{E}_v^\xi)$ that encapsulates the localized higher-order interaction. The node set $\mathcal{V}_v^\xi$ consists of the center node $\{v\}$ and its non-center neighbors $\mathcal{N}_{nc}(v) = \{v' \mid v' \in \mathcal{N}(v) \wedge \Phi(v') \neq o_c\}$, while the edge set $\mathcal{E}_v^\xi = \{(v, u) \mid u \in \mathcal{N}_{nc}(v)\}$ defines the internal relations.

While the definition of HOU's is grounded in star-shaped schemas, real-world heterogeneous networks that exhibit multi-center patterns can also be unified under this framework by treating each connected group of center nodes (one per center type) as a composite anchor. For such networks, a HOU is defined as a subgraph that includes all center nodes from a group together with their shared non-center neighbors. This unified HOU captures the joint event where multiple center entities co-occur through common functional associations. Fig. 3(b) illustrates a typical multi-center pattern, where Gene and Compound jointly serve as centers, and their neighbors (e.g., Disease, Species) complete the HOU.

We assume a heterogeneous network H is generated by a hierarchical process: (i) sample center nodes $\mathcal{V}_c$; (ii) each center node independently generates its non-center neighbors, forming HOU's; (iii) the global skeleton emerges from HOU interactions via shared non-center nodes. HOU's serve a dual purpose: internally they preserve pairwise relations like hyperedges; externally they interconnect to form the network backbone. In this paper, our REHHOR does not directly maximize the likelihood of H . Instead, it instantiates HOU structure and optimizes a denoising objective plus a contrastive loss, which can be interpreted as a variational lower bound of the log-likelihood of observed HOU interactions. Thus, REHHOR is a hybrid model: its architecture encodes the structural prior of the generative process, while its diffusion-based learning provides generative capacity to recover missing structures. This combination ensures robustness under data sparsity.

3.3. Relation prediction tasks

REHHOR aims to enhance relation prediction in heterogeneous networks by modeling the higher-order generative process. Given a potentially incomplete heterogeneous network $H = (\mathcal{V}, \mathcal{E}, \Phi, \Psi, \mathcal{T}, \mathcal{T}_\mathcal{E})$, we define three hierarchical prediction tasks to validate our framework.

Definition 3.6 (Node Relation Prediction.). The objective is to learn a mapping function $f_e : \mathcal{V} \times \mathcal{V} \rightarrow \{0, 1\}$ that determines the existence of an edge between any node pair (v_i, v_j) . Specifically, $f_e(v_i, v_j) = 1$ if $(v_i, v_j) \in \mathcal{E}$, and 0 otherwise.

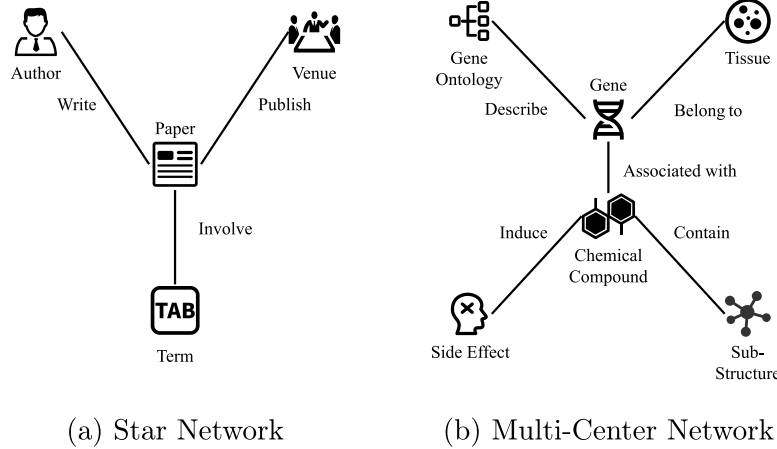


Fig. 3. Different patterns of heterogeneous networks.

Definition 3.7 (Type-Specific Relation Prediction.). For a target relation type $t \in \mathcal{T}_E$, the goal is to learn a function f_t^i such that $f_t^i(v_i, v_j) = 1$ if $(v_i, v_j) \in \mathcal{E} \wedge \Psi((v_i, v_j)) = t$, and 0 otherwise.

Definition 3.8 (HOU Interaction Relation Prediction.). For a specific non-center node type o , the task is to learn a function f_ξ^o such that $f_\xi^o(\xi_{v_i}, \xi_{v_j}) = 1$ if the higher-order units (ξ_{v_i}, ξ_{v_j}) share at least one node v_k of type o , signifying that $v_k \in \mathcal{V}_{v_i}^\xi \cap \mathcal{V}_{v_j}^\xi \wedge \Phi(v_k) = o$. Otherwise, $f_\xi^o(\xi_{v_i}, \xi_{v_j}) = 0$.

The selection of these three hierarchical tasks provides a multi-granular evaluation of our framework. Node relation prediction establishes a baseline for general predictive capability, while type-specific relation prediction assesses the model's ability to resolve semantic homogenization by distinguishing multifaceted entity interactions. Finally, HOU interaction relation prediction serves to verify the reconstruction of the global interaction skeleton. Collectively, these tasks validate the model's proficiency in capturing the entire hierarchical generation process, spanning from fine-grained node associations to the macro-level structural backbone.

4. Methodology

This section describes the four primary stages of the framework: HOU extraction, bipartite graph construction, hierarchical anisotropic encoding, and HOU interaction relation reconstruction.

The overall architecture of REHHOR framework is illustrated in Fig. 4. The model follows a pipeline comprising four primary stages. (1) HOU Extraction: This module identifies HOUs and their underlying relations from the observed heterogeneous network to restore the hierarchical generation process; (2) Node-HOU Bipartite Graph Construction: This module distinguishes the internal relations of these HOUs based on node types, thereby capturing structural heterogeneity for subsequent encoding. (3) Hierarchical Anisotropic Encoding: This module iteratively learns node and HOU representations through encoding and pooling processes guided by a directed diffusion mechanism. (4) HOU Interaction Relation Reconstruction: This Module focuses on the explicit recovery of the interaction skeleton, ensuring that the learned embeddings effectively preserve the network backbone.

4.1. Extraction of higher-order units

As defined in Definition 3.5, a HOU within a heterogeneous network $H = (\mathcal{V}, \mathcal{E}, \Phi, \mathcal{T})$ is a subgraph ξ_v consisting of a center node v with $\Phi(v) = o_c$ and its set of non-central neighbors $\mathcal{N}_{nc}(v)$. REHHOR extracts the complete set of HOUs in the network as follows:

$$\Xi = \{\xi_v \mid v \in \mathcal{V} \wedge \Phi(v) = o_c\}, \quad (1)$$

The extracted set Ξ serves as the structural foundation for both the encoding and optimization stages, enabling the model to explicitly simulate the hierarchical generation process of the heterogeneous network. The HOU extraction is not intended to discover unknown latent structures but to materialize the assumed generative anchors, providing a strong structural prior for the downstream encoder.

4.2. Node-HOU bipartite graph construction

Each HOU contains several internal relations of different types. Representing these interactions through a single hyperedge often leads to the loss of critical heterogeneous information. To mitigate this, some hypergraph-based methods partition the network into multiple homogeneous components based on edge types or meta-paths. However, this approach fails to explicitly capture the integrated characteristics within an HOU. To balance structural heterogeneity with information integrity, REHHOR treats each HOU as a distinct node and extracts its internal relations according to node types. This process results in the construction of several bipartite graphs, each composed of a specific node type and the set of HOU nodes.

For a specific node type $o \in \mathcal{T}$, the corresponding bipartite graph is defined as $B_o = (\mathcal{V}_o^B, \mathcal{E}_o^B)$, where $\mathcal{V}_o^B = \Xi \cup \mathcal{V}_o$ and $\mathcal{E}_o^B = \{(v', \xi_v) \mid v' \in \mathcal{V}_v^\xi \wedge \Phi(v') = o \wedge \xi_v \in \Xi\}$. Here, $\mathcal{V}_o = \{v \mid v \in \mathcal{V} \wedge \Phi(v) = o\}$ represents the complete set of nodes of type o . The structural dependencies within B_o are encoded into an adjacency matrix $\mathbf{B}^o \in \{-1, 1\}^{|\mathcal{V}_o| \times |\Xi|}$. $\mathbf{B}_{ij}^o$ is set to 1 if the i th node v_i is contained within the j th HOU, and -1 otherwise. Notably, we employ -1 instead of the conventional 0 for two reasons. First, the signed encoding centers the representation space (zero mean for balanced memberships), which improves numerical stability and makes the column-wise statistics μ and σ meaningful. They directly capture the tendency of each HOU to include or exclude nodes of type o . Second, the sign operation in the directed diffusion mechanism is well-defined and semantically natural when entries are ± 1 , enabling the noise to align with the original membership direction. This design choice is not an ad hoc trick but a deliberate component that facilitates anisotropic noise injection and symmetric geometry, both critical for preserving structural signals during the generative process.

4.3. Hierarchical anisotropic encoding

REHHOR initiates the encoding process by learning representations for nodes and HOUs within each bipartite graph B_o . Given the complexity of interaction distributions between entities, we employ a denoising diffusion framework, which has demonstrated superior capacity for capturing intricate data distribution through a progressive noise-addition and reverse-denoising process (Ho et al., 2020). However, conventional

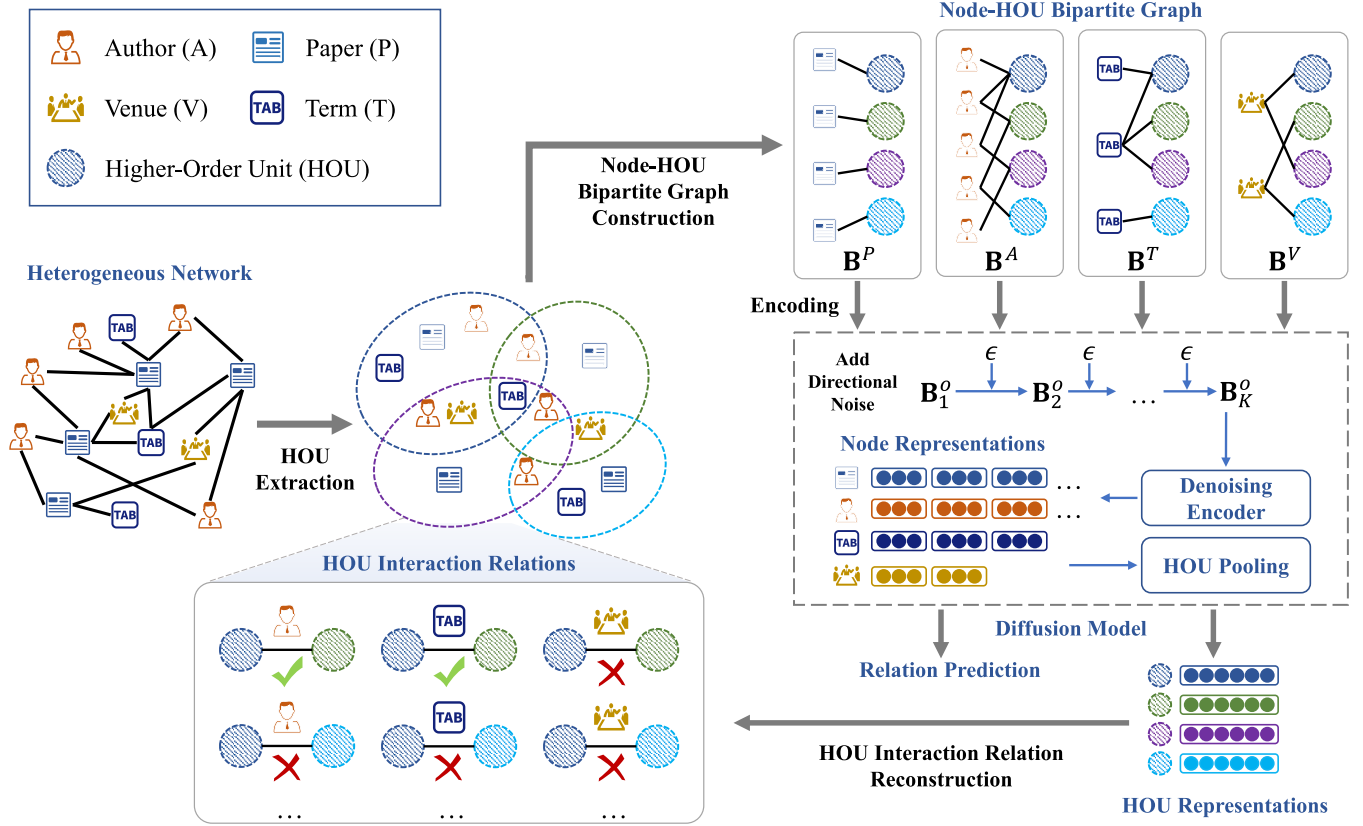


Fig. 4. Framework diagram of the REHHOR model.

diffusion models rely on isotropic Gaussian noise, which, as previously evidenced by the PCA results in Fig. 2, fails to account for the inherent anisotropy of heterogeneous networks. This mismatch often leads to excessive signal dilution, where the structural signals are rapidly overwhelmed by noise, thereby hindering the training of an effective denoising encoder.

To address this challenge, REHHOR incorporates a directed diffusion mechanism that injects data-dependent, directional noise during the forward process (Yang et al., 2023a). For a node v_i at the k th diffusion step, the noise addition is formulated as follows:

$$\mathbf{B}_{i,k}^o = \sqrt{\alpha_k} \mathbf{B}_{i,0}^o + \sqrt{1 - \alpha_k} \epsilon_o', \quad (2)$$

$$\epsilon_o' = \text{sgn}(\mathbf{B}_{i,0}^o) \odot \bar{\epsilon}_o, \quad (3)$$

$$\bar{\epsilon}_o = \boldsymbol{\mu}_o + \boldsymbol{\sigma}_o \odot \epsilon_o, \quad (4)$$

where $\mathbf{B}_{i,0}^o = \mathbf{B}_i^o$ is the corresponding vector representing the relation of node v_i in the original adjacency matrix. $\boldsymbol{\mu}_o \in \mathbb{R}^{|\Xi|}$ is a vector whose j th component is the mean of the j th column across all nodes of type o :

$$\boldsymbol{\mu}_o[j] = \frac{1}{|\mathcal{V}_o|} \sum_{i=1}^{|\mathcal{V}_o|} \mathbf{B}_{ij}^o. \quad (5)$$

Similarly, $\boldsymbol{\sigma}_o \in \mathbb{R}^{|\Xi|}$ is a vector whose j th component is the standard deviation of the j th column:

$$\boldsymbol{\sigma}_o[j] = \sqrt{\frac{1}{|\mathcal{V}_o|} \sum_{i=1}^{|\mathcal{V}_o|} (\mathbf{B}_{ij}^o - \boldsymbol{\mu}_o[j])^2}. \quad (6)$$

$\odot$ is the Hadamard product. ϵ_o is noise sampled from the standard Gaussian distribution $\text{Gaussian}(\mathbf{0}, \mathbf{I})$. $\bar{\alpha}_k = \prod_{i=0}^k (1 - \beta_i) \in (0, 1)$ is calculated according to the variance list for adding noise in the denoising diffusion model (Ho et al., 2020) which is a descending sequence $\beta_1, \dots, \beta_K$ with elements in $(0, 1)$. Unlike standard isotropic models, Eq. (4) transforms data-independent noise into an anisotropic distribution tailored

to the data characteristics, while Eq. (3) ensures that the added noise aligns with the direction of the original adjacency signals. This strategy prevents rapid signal degradation and ensures that the subsequent denoising encoder learns robust, structurally-aware representations.

To denoise the perturbed adjacency matrix and extract robust node representations, REHHOR employs a multi-layer perceptron (MLP) with residual connections as the denoising encoder. This encoder maps the learned features into a d -dimensional space through a final linear layer, a process formally expressed as:

$$\hat{\mathbf{z}}_{i,o}^{(l)} = \text{MLP}_{\text{enc}}^{(l)}(\hat{\mathbf{z}}_{i,o}^{(l-1)}) + \hat{\mathbf{z}}_{i,o}^{(l-1)}, 0 < l \leq L, \quad (7)$$

$$\mathbf{z}_{i,o} = \text{Linear}(\hat{\mathbf{z}}_{i,o}^{(L)}), \quad (8)$$

where $\text{MLP}_{\text{enc}}^{(l)}$ denotes the l th layer of the perceptron. The initial input is the noisy adjacency vector from the final diffusion step, such that $\hat{\mathbf{z}}_{i,o}^{(0)} = \mathbf{B}_{i,K}^o$. Once the type-specific node representations are obtained from the respective bipartite graphs, REHHOR utilizes a pooling operation to extract HOU representations. For each HOU $\xi_{v_i} \in \Xi$ on the bipartite graph B_o , the representation is computed as:

$$\hat{\mathbf{h}}_{i,o} = [\mathbf{z}_{i,o_c}, \sum_{v_j \in \mathcal{N}(v_i)} \mathbf{z}_{j,o}], \quad (9)$$

$$\mathbf{h}_{i,o} = \text{MLP}_{\text{pool}}(\hat{\mathbf{h}}_{i,o}), \quad (10)$$

where $\mathbf{z}_{i,o_c}$ is the representation of the center node v_i within B_{o_c} , and o represents a non-central node type. The term $[\cdot, \cdot]$ denotes the concatenation operation. The pooling MLP_{pool} is shared across all non-central node types to promote parameter efficiency. In this pooling mechanism, the center node representation $\mathbf{z}_{i,o_c}$ acts as a structural anchor for its neighbors. For non-central nodes, the summation-based aggregation is employed to maximally preserve neighborhood information while remaining parameter-free (Xu et al., 2019).

4.4. HOU interaction relation reconstruction

Standard diffusion models are typically trained by optimizing the variational lower bound of the reverse denoising process. However, deriving a closed-form posterior distribution under REHHOR's directed noise injection is analytically intractable. To circumvent this, we draw inspiration from empirical solutions in recent diffusion research (Bansal et al., 2024; Li et al., 2022b) that prioritize the direct reconstruction of pre-noised features to adaptively mitigate noise interference. Diverging from conventional graph generative models that reconstruct individual edges or meta-paths (Tian et al., 2023), REHHOR focuses on reconstructing the interaction relations between HOU to capture the overarching backbone of the heterogeneous network. For each non-central node type o , we optimize the following contrastive loss:

$$\mathcal{J}_o = \sum_{\xi_{v_i} \in \Xi} \left(-\log \sum_{\xi_{v_j} \in \Xi_{i,\text{pos}}^o} \exp(\text{sim}(\mathbf{h}_{i,o}, \mathbf{h}_{j,o})/\tau) + \log \sum_{\xi_{v_n} \in \Xi} \exp(\text{sim}(\mathbf{h}_{i,o}, \mathbf{h}_{n,o})/\tau) \right), \quad (11)$$

where τ denotes the temperature parameter and $\text{sim}(\cdot, \cdot)$ represents cosine similarity. The positive sample set $\Xi_{i,\text{pos}}^o$ consists of HOU connected to ξ_{v_i} via nodes of type o , defined as:

$$\Xi_{i,\text{pos}}^o = \{\xi_{v_j} | \xi_{v_j} \in \Xi - \{\xi_{v_i}\} \wedge \mathcal{V}_o \cap \mathcal{V}_{v_i}^{\xi} \cap \mathcal{V}_{v_j}^{\xi} \neq \emptyset \wedge o \neq o_c\}, \quad (12)$$

This objective extends the InfoNCE loss (Oord et al., 2018) to the scale of HOU, aiming to discriminate between genuine HOU interactions and noise-driven associations. To enhance computational efficiency, we employ a negative sampling strategy by uniformly selecting a fixed number of non-adjacent HOU to form the negative set $\Xi_{i,\text{neg}}^o \subset \Xi - \Xi_{i,\text{pos}}^o$. Consequently, the loss function in Eq. (11) is approximated as:

$$\mathcal{J}_o = \sum_{\xi_{v_i} \in \Xi} \left(-\log \sum_{\xi_{v_j} \in \Xi_{i,\text{pos}}^o} \exp(\text{sim}(\mathbf{h}_{i,o}, \mathbf{h}_{j,o})/\tau) + \log \sum_{\xi_{v_n} \in \Xi_{i,\text{neg}}^o} \exp(\text{sim}(\mathbf{h}_{i,o}, \mathbf{h}_{n,o})/\tau) \right), \quad (13)$$

Theoretically, using a limited number of negative samples can obtain the same result as the original loss function, but in some cases, using the negative sampling strategy can obtain better results than using the entire sample set. While negative sampling is stochastic, it inherently favors the selection of structurally distant HOU, thereby reinforcing the model's ability to distinguish the global backbone from disconnected components.

Finally, REHHOR jointly optimizes the interaction relations across all non-central node types:

$$\mathcal{J} = \sum_{o \in \mathcal{T} - \mathcal{T}_c} \mathcal{J}_o, \quad (14)$$

This joint optimization ensures that HOU representations effectively characterize the network backbone. Furthermore, through the coupled encoding of central nodes and neighboring non-central nodes in formula (10), the node representations can finely fit the local relation under the backbone structure.

The pseudocode of REHHOR is shown in Algorithm 1.

4.5. Complexity analysis

The computational efficiency of REHHOR is analyzed across its three primary phases. First, the construction of Node-HOU bipartite graphs requires a single traversal of the edge set, resulting in a time complexity of $O(|\mathcal{E}|)$. Second, the generation of d -dimensional representations for nodes and HOU through the directed diffusion process scales as $O(|\mathcal{V}|d)$. Finally, the reconstruction optimization involves HOU sampling and contrastive loss computation. Given that the number of HOU satisfies $|\Xi| < |\mathcal{V}|$, the sampling process remains efficient. Since the calculation of the contrastive loss fundamentally entails traversing the

structural dependencies within the network, its complexity is $O(|\mathcal{E}|d)$. Consequently, the overall time complexity of REHHOR maintains a linear relationship with the number of edges $|\mathcal{E}|$, demonstrating superior scalability for large-scale heterogeneous networks.

Algorithm 1 REHHOR ($H, \{\beta_1, \dots, \beta_K\}, d, \tau$).

Input: Heterogeneous network H , variance schedule $\{\beta_1, \dots, \beta_K\}$, representation dimension d , temperature parameter τ of contrastive loss;

Output: Node representations $\{\mathbf{z}_{i,o} \in \mathbb{R}^d\}_{v_i \in \mathcal{V}}$, HOU representations $\{\mathbf{h}_{i,o} \in \mathbb{R}^d\}_{\xi_{v_i} \in \Xi}$;

- 1: Extract HOU set Ξ ; ▷ See Definition 3.5
 - 2: Construct positive/negative samples $\Xi_{i,\text{pos}}^o$ and $\Xi_{i,\text{neg}}^o$; ▷ See Eq. (12);
 - 3: Construct node-HOU bipartite graph $\{\mathcal{B}_o\}_{o \in \mathcal{T}}$ from heterogeneous network H ;
 - 4: **for** $o \in \mathcal{T}$ **do**
 - 5: Add directed noise to the bipartite graph adjacency matrix; ▷ See Eq. (2)–(4);
 - 6: Apply encoder to generate node representations $\{\mathbf{z}_{i,o}\}_{v_i \in \mathcal{V}}$; ▷ See Eq. (7)–(8);
 - 7: Apply pooling operation to obtain HOU representations $\{\mathbf{h}_{i,o}\}_{\xi_{v_i} \in \Xi}$; ▷ See Eq. (10);
 - 8: **end for**
 - 9: Training model based on skeletal contrastive loss; ▷ See Eq. (13);
 - 10: **return** Node representations $\{\mathbf{z}_{i,o}\}_{v_i \in \mathcal{V}}$, HOU representations $\{\mathbf{h}_{i,o}\}_{\xi_{v_i} \in \Xi}$
-

5. Theoretical insights of directed diffusion mechanism

This section provides a theoretical analysis of the directed diffusion mechanism, including its preservation of anisotropy, convergence properties, and relationship to cold diffusion and denoising autoencoders.

The directed diffusion mechanism of REHHOR follows the same multi-step forward process as DDPMs (Ho et al., 2020) but differs in two key aspects: (i) we inject directional, data-dependent noise instead of isotropic Gaussian noise to preserve anisotropic structures; (ii) we train the denoising encoder to directly reconstruct the original HOU interactions via a contrastive loss, rather than predicting the noise. This design retains the benefits of multi-scale denoising while aligning the objective with our downstream task of relation prediction. In the following, we provide a detailed theoretical analysis.

Sign Alignment Preserves Anisotropy. In standard DDPMs, isotropic Gaussian noise $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ is added independently of the data. For an anisotropic data distribution concentrated along certain directions, isotropic noise quickly “washes out” the directional structure because the noise variance is uniform in all directions. In contrast, our directed noise is defined as:

$$\epsilon' = \text{sgn}(x_0) \odot (\mu + \sigma \odot \epsilon),$$

where x_0 is the original adjacency vector (with entries ± 1). The sign alignment constraint $\text{sgn}(x_0)$ ensures that the added noise vector points in the same half-space as the original data point for each coordinate. This prevents the noise from flipping the sign of the signal, which would otherwise cause rapid cancellation of directional information. More formally, let $s = \text{sgn}(x_0)$. The noisy observation at step k is $x_k = \sqrt{\bar{\alpha}_k} x_0 + \sqrt{1 - \bar{\alpha}_k} \epsilon'$. The expected inner product between x_k and x_0 is:

$$\mathbb{E}[x_k^T x_0] = \sqrt{\bar{\alpha}_k} \|x_0\|^2 + \sqrt{1 - \bar{\alpha}_k} \mathbb{E}[(s \odot (\mu + \sigma \odot \epsilon))^T x_0].$$

Since $x_0 = s \odot |x_0|$, we have $(s \odot (\mu + \sigma \odot \epsilon))^T x_0 = |x_0|^T (\mu + \sigma \odot \epsilon)$. The terms μ and σ are computed from the data distribution, ensuring that the noise has non-negative correlation with $|x_0|$ in expectation. Consequently, the signal-to-noise ratio decays more slowly along the principal directions of the data, preserving anisotropy.

Convergence of the Forward Process. The forward process with directed noise does **not** converge to a simple isotropic Gaussian

distribution. Instead, it converges to a data-dependent anisotropic distribution. As $k \rightarrow \infty$, $\sqrt{\tilde{\alpha}_k} \rightarrow 0$ and $\sqrt{1 - \tilde{\alpha}_k} \rightarrow 1$. Hence, $x_\infty = \epsilon'$, where $\epsilon' = s \odot (\mu + \sigma \odot \epsilon)$ with $\epsilon \sim \mathcal{N}(0, \mathbf{I})$. This limiting distribution is a mixture of scaled and sign-perturbed Gaussians, parameterized by the empirical moments μ and σ computed from the training data. Importantly, this stationary distribution retains the anisotropic structure of the original data because μ and σ vary across dimensions, and the sign alignment preserves the directional bias.

Relation to Cold Diffusion and Denoising Autoencoders. Standard diffusion models derive a variational lower bound that relies on a closed-form Gaussian posterior and a fixed isotropic prior. For our directed noise, the true reverse posterior is not Gaussian, and a strict VLB with a fixed prior is unavailable. However, this does not leave the method without theoretical grounding. The cold diffusion framework (Bansal et al., 2024) demonstrates that one can train a network to directly predict the original signal x_0 from the corrupted x_k using the simple MSE loss $\|f_\theta(x_k, k) - x_0\|^2$, without requiring a closed-form posterior or a fixed prior. This loss is exactly what we optimize (plus the contrastive term). Our directed noise injection is a particular instantiation of the forward process in cold diffusion, tailored to preserve anisotropic structure. Additionally, the same objective can be interpreted as a denoising autoencoder (DAE) (Vincent et al., 2010), for which it is known that minimizing the reconstruction error on corrupted inputs approximates the score of the data distribution. The directional noise ensures that the corruption does not destroy anisotropic signals, making the score estimation more accurate for downstream relation prediction. Thus, while we do not claim a novel generative bound, our training objective is firmly grounded in established representation learning theory.

Impact on Reverse Diffusion Optimality. The optimal reverse diffusion process is defined by the true posterior $q(x_{t-1}|x_t, x_0)$. For our directed noise, this posterior is not available in closed form due to the sign and scale transformations. However, by training the encoder to directly reconstruct x_0 (or equivalently, the HOU interaction structure) via the contrastive loss, we bypass the need to explicitly model the reverse transition. This design choice is optimal for our discriminative objective: we do not require a generative reverse process that can sample from $p_\theta(x_{t-1}|x_t)$; we only need a representation from which the original skeleton can be recovered. The direct reconstruction objective can be viewed as a form of “implicit” reverse process that is tailored to the downstream task. Empirically, this approach achieves superior performance under sparsity (see ablation study, Fig. 6), indicating that the trade-off-sacrificing exact generative modeling for task-specific reconstruction-is beneficial.

6. Experimental verification

To evaluate the effectiveness of REHHOR, we conduct experiments on four real-world datasets across three relation prediction tasks, comparing with 14 baseline methods.

6.1. Real-world benchmark datasets

To evaluate the effectiveness of REHHOR, relation prediction experiments are conducted on three real-world datasets spanning diverse domains. These publicly available benchmark datasets, originally sourced from Yang et al. (2022), provide a comprehensive basis for testing model performance across different network topologies. The basic statistics of these datasets are summarized in Table 2, and their specific characteristics are detailed below.

- **DBLP.**¹ Sourced from the computer science bibliographic database, this academic network comprises 4057 authors (A), 14,328 papers

(P), 20 conferences (V), and 7723 terms (T). It contains a total of 119,783 edges representing multifaceted scholarly interactions, where papers (P) function as the central node type.

- **PubMed.**² This biomedical co-occurrence network originates from the PubMed database and focuses on molecular and life sciences. The network incorporates 26,522 chemical substances (C), 13,561 genes (G), 2863 species (S), and 20,163 diseases (D), totaling 60,394 edges. Within this structure, genes (G) are designated as the central node type.
- **Yelp.**³ Derived from the Yelp business review platform, this large-scale network characterizes commercial ecosystems. It includes 39 locations (L), 7474 businesses (B), 9 star ratings (S), and 74,943 reviews (P). With 3,996,417 edges, primarily distributed between businesses and review phrases, this dataset highlights dense interaction patterns. Businesses (B) serve as the central node type in this context.
- **Hetionet (Multi-Center Subset).**⁴ This biomedical knowledge graph subset is extracted from Hetionet, a heterogeneous network integrating diverse biomedical entities. The extracted subgraph comprises six node types: genes (G), compounds (C), side effects (S), diseases (D), pathways (P), and anatomy (A). It contains 19,145 genes, 1538 compounds, 136 diseases, 5701 side effects, 1822 pathways, and 400 anatomy terms, with a total of about 729,000 edges. This network exhibits a multi-center pattern where connected G and C jointly serve as central node types, forming a minimal center set $\{G, C\}$ that covers all node types together with their neighbors.

6.2. Baseline methods

To comprehensively evaluate the performance of REHHOR, we select a wide range of representative homogeneous and heterogeneous network representation learning methods as baselines. These methods encompass various techniques including random walks, matrix factorization, and deep learning.

(1) Homogeneous Network Representation Learning. When applying these methods, node and edge types are ignored to treat the heterogeneous network as a homogeneous graph.

- **node2vec** (Grover & Leskovec, 2016). This method extends DeepWalk by introducing a tunable random walk strategy that balances breadth-first and depth-first searches to capture diverse node proximity information. (Complexity: $O(|V|)$)
- **LINE** (Tang et al., 2015b). By modeling first-order and second-order similarities, LINE directly optimizes node pair relations and employs edge sampling to achieve efficient low-dimensional representations for large-scale graphs. (Complexity: $O(|E|)$)

(2) Heterogeneous Network Representation Learning. These methods explicitly account for the heterogeneity of nodes and relations through various specialized architectures.

- **TransE** (Bordes et al., 2013). This foundational knowledge graph embedding method models relations as vector translations between entities and optimizes representations via triplet-based scoring. (Complexity: $O(|E|)$)
- **PTE** (Tang et al., 2015a). This approach decomposes heterogeneous networks into multiple bipartite graphs based on edge types and learns node embeddings by reconstructing first-order neighbor relations. (Complexity: $O(|E|)$)
- **metapath2vec** (Dong et al., 2017). Utilizing meta-path-based random walks, this model generates node sequences that capture heterogeneous semantics, which are then processed via Skip-gram to learn representations. (Complexity: $O(|V|)$)

² <https://www.ncbi.nlm.nih.gov/pubmed>

³ <https://www.yelp.com/dataset/challenge>

⁴ <https://het.io/>

¹ <https://dblp.org>

Table 2
Statistical information of used datasets.

Dataset	Number of Nodes		Number of Edges	
DBLP	A: 4,057	V: 20	P-A: 19,645	P-T: 85,810
	P: 14,328	T: 7,723	P-V: 14,328	
PubMed	C: 2,652	S: 2,863	G-C: 31,277	G-S: 3,155
	G: 13,561	D: 20,163	G-D: 25,962	
Yelp	L: 39	S: 9	B-L: 7,474	B-S: 7,474
	B: 7,474	P: 74,943	B-P: 3,981,469	
Hetionet	G: 19,145	C: 1,538	G-C: 18,756	C-S: 138,944
	D: 136	S: 5,701	G-P: 84,372	
	P: 1,822	A: 400	G-A: 526,407	C-D: 755

- **HIN2vec** (Fu et al., 2017). This method learns node representations by predicting the existence of specific meta-paths between node pairs, effectively capturing proximity under different structural constraints. (Complexity: $O(|\mathcal{V}|)$)
- **ConvE** (Dettmers et al., 2018). By employing 2D convolutional neural networks to model triplets, ConvE achieves higher expressiveness compared to traditional geometric distance-based methods. (Complexity: $O(|\mathcal{E}|)$)
- **JUST** (Hussein et al., 2018). This model optimizes random walks through jump-and-stay strategies, which eliminates the requirement for predefined meta-paths while balancing the distribution of node types. (Complexity: $O(|\mathcal{V}|)$)
- **R-GCN** (Schlichtkrull et al., 2018). Extending graph convolutional networks, R-GCN models relational heterogeneity by learning type-specific weight matrices to capture the complex structure of various interactions. (Complexity: $O(|\mathcal{E}|)$)
- **HGT** (Hu et al., 2020). This Transformer-inspired architecture utilizes a self-attention mechanism to learn node representations by reconstructing triplets composed of diverse node and edge types. (Complexity: $O(|\mathcal{V}|)$)
- **CPT-HG** (Jiang et al., 2021b). This contrastive pre-training framework designs edge-level and subgraph-level tasks with advanced negative sampling strategies to enhance representation robustness. (Complexity: $O(|\mathcal{E}|)$)
- **HeteHG-VAE** (Fan et al., 2021). By treating the network as a hypergraph, this model employs variational autoencoders and attention mechanisms to reconstruct relations between nodes and hyperedges. (Complexity: $O(|\mathcal{E}|)$)
- **NHGM** (Li et al., 2024). This method optimizes contrastive learning through a mutual information maximization strategy, generating multiple views based on relations and meta-paths. (Complexity: $O(|\mathcal{E}|)$)
- **MEOW** (Yu et al., 2024). This framework utilizes multi-granularity enhanced views and GCNs, incorporating weighted negative sampling to improve the effectiveness of contrastive learning. (Complexity: $O(|\mathcal{E}|)$)

Here we report the time complexities of these methods as functions of the input network statistics (i.e., number of nodes $|\mathcal{V}|$ and edges $|\mathcal{E}|$) for reference. Nevertheless, we caution that evaluating efficiency solely on the basis of these asymptotic complexities can be problematic. The methods vary considerably in their hyperparameter settings, learning paradigms, and training strategies, which hinders the establishment of a unified comparative standard.

Among the aforementioned methods, HeteHG-VAE attempts to capture higher-order structures by constructing hyperedges, while the remaining baselines focus exclusively on capturing or reconstructing pairwise relations between nodes. However, even HeteHG-VAE fails to depict the comprehensive hierarchical generation process of heterogeneous networks.

6.3. Experimental setup and environment

REHHOR is evaluated across three primary tasks: node relation prediction, type-specific node relation prediction, and HOU interaction relation prediction. While the first two assess the model's ability to recover micro-level connectivity, the third specifically validates the reconstruction of the macro-level interaction skeleton. All tasks are formulated as binary classification problems, where we predict the existence of relations between node pairs or HOUs. To quantify performance, we employ AUC (Area Under the ROC Curve) and AP (Average Precision) as evaluation metrics. AUC measures the probability that a randomly chosen positive sample receives a higher score than a negative one, calculated as follows:

$$\text{AUC} = \frac{\sum_{i \in D_{\text{pos}}} \sum_{j \in D_{\text{neg}}} \mathbb{I}(p(i) > p(j))}{|D_{\text{pos}}| \cdot |D_{\text{neg}}|}, \quad (15)$$

where D_{pos} and D_{neg} denote the positive and negative sample sets, respectively, and $\mathbb{I}$ is the indicator function. Complementarily, AP evaluates the precision-recall trade-off by calculating the area under the precision-recall curve:

$$\text{AP} = \frac{1}{|D_{\text{pos}}|} \sum_{k=1}^{|D_{\text{pos}}|} \text{Precision}(k) \cdot \text{pos}(k), \quad (16)$$

where k represents the rank in the prediction sequence, and $\text{pos}(k)$ indicates whether the k th sample is a positive example. Both metrics range from 0 to 1, with higher values signifying superior predictive accuracy.

Data preprocessing and edge deletion. For each dataset, we randomly remove edges from each relation type independently (e.g., P-A, P-V, P-T in DBLP) to preserve the original type distribution. The deletion ratios are set to 20%, 40%, 60%, and 80% of the edges per relation type, ensuring that no relation type becomes completely absent under high sparsity. The remaining edges constitute the training network. The removed edges are held out as positive test samples. We also sample an equal number of non-existent edges from the original network as negative test samples, matching the type distribution of positive samples where applicable. For each deletion ratio, we perform a single random split: 80% of the removed edges are used as the test set, and the remaining 20% of removed edges are used as a validation set for early stopping and hyperparameter selection. This split is performed after the deletion step, so the validation set is never seen during training. For each configuration, we run five independent trials with different random seeds and report the average AUC/AP with standard deviations.

Model optimization and hyperparameters. For REHHOR, we employ the Adam stochastic gradient optimizer across all datasets for a total of 50 epochs, with the learning rate set to 0.01. The denoising encoder and the pooling module are both implemented as a 3-layer MLP with ReLU activation. The temperature parameter τ in the contrastive loss is set to 0.9 for DBLP and 0.5 for Yelp and PubMed. To ensure a fair comparison, the representation dimension d is fixed to 128 for all

Table 3

Results of node relation prediction in the DBLP network (%).

Deleted Edges	20%		40%		60%		80%	
	AUC	AP	AUC	AP	AUC	AP	AUC	AP
node2vec	70.3±2.1	68.1±1.9	68.5±2.4	65.3±2.2	63.7±1.8	61.0±1.7	61.5±2.3	58.1±2.0
LINE	75.1±1.6	73.2±1.4	74.4±1.9	71.9±1.7	72.7±2.2	69.7±1.8	67.6±1.5	64.8±1.3
TransE	75.0±2.5	73.1±2.3	69.8±2.0	68.1±1.8	60.5±2.7	58.2±2.4	50.1±1.9	48.3±1.7
PTE	54.6±1.3	52.8±1.1	54.5±1.6	52.6±1.4	53.6±1.2	51.5±1.0	53.0±1.8	51.4±1.5
metapath2vec	73.3±2.8	66.8±2.5	66.1±2.2	59.8±2.0	60.1±2.4	55.5±2.1	54.6±1.7	52.4±1.4
HIN2vec	70.9±1.8	67.4±1.6	65.1±2.1	61.8±1.9	71.8±1.5	69.5±1.3	53.9±2.2	50.0±1.9
ConvE	71.2±1.4	71.5±1.2	70.5±1.7	70.6±1.5	61.6±1.9	62.9±1.6	63.2±2.1	62.9±1.8
JUST	48.9±1.1	47.5±0.9	49.2±1.4	48.7±1.2	49.1±1.0	49.3±0.8	48.7±1.3	49.3±1.1
R-GCN	49.0±0.9	47.2±0.8	49.1±1.2	47.5±1.0	49.2±1.4	47.5±1.1	49.0±1.6	47.2±1.3
HGT	60.5±2.0	63.3±1.7	54.2±1.8	57.5±1.5	48.7±2.2	52.4±1.9	47.0±1.4	49.4±1.2
CPT-HG	<u>84.1±1.3</u>	<u>81.3±1.1</u>	79.4±1.6	76.8±1.4	76.6±1.2	74.8±1.0	72.8±1.8	69.4±1.5
HeteHG-VAE	82.6±1.0	<u>81.3±0.9</u>	<u>82.3±1.1</u>	<u>81.2±0.9</u>	<u>81.8±1.3</u>	<u>81.1±1.1</u>	<u>80.1±1.2</u>	<u>80.2±1.0</u>
NHGMI	82.4±1.5	79.2±1.3	76.9±1.7	74.1±1.4	74.6±1.1	73.1±0.9	69.6±1.6	66.5±1.3
MEOW	81.8±1.2	77.7±1.0	76.5±1.4	73.4±1.2	74.0±1.6	72.0±1.3	68.8±1.1	66.2±0.9
REHHOR	84.3±0.9*	83.0±1.1*	83.5±1.0*	82.3±0.8*	83.0±1.2*	81.6±0.9*	82.6±1.0*	81.5±1.1*

Table 4

Results of node relation prediction in the PubMed network (%).

Deleted Edges	20%		40%		60%		80%	
	AUC	AP	AUC	AP	AUC	AP	AUC	AP
node2vec	74.0±2.2	71.5±2.0	72.4±1.8	70.1±1.6	70.3±2.4	68.2±2.1	66.6±1.9	64.2±1.7
LINE	75.8±1.5	78.3±1.3	75.0±2.1	76.4±1.8	<u>72.8±1.7</u>	73.3±1.5	68.6±2.2	67.8±1.9
TransE	68.6±2.7	73.3±2.4	67.3±2.0	74.2±1.8	63.2±2.5	72.7±2.2	55.6±1.6	66.3±1.4
PTE	51.6±1.2	55.6±1.0	51.4±1.4	55.9±1.2	51.4±1.8	56.4±1.5	51.1±1.1	57.3±0.9
metapath2vec	78.5±1.9	76.3±1.7	75.8±2.3	72.8±2.0	72.0±1.4	69.3±1.2	70.8±2.1	67.1±1.8
HIN2vec	62.2±2.5	52.7±2.2	47.1±2.8	54.0±2.4	47.4±2.0	53.3±1.7	66.6±1.5	62.4±1.3
ConvE	73.0±1.6	73.8±1.4	74.0±1.9	74.2±1.6	72.2±2.2	71.2±1.9	68.3±1.8	66.2±1.5
JUST	48.6±1.0	51.3±0.8	49.1±1.3	52.6±1.1	49.3±1.6	53.6±1.3	48.9±0.9	53.0±0.7
R-GCN	48.9±1.4	47.3±1.2	49.2±1.7	47.5±1.4	48.8±1.1	47.4±0.9	49.7±2.0	48.2±1.7
HGT	68.9±2.1	75.3±1.8	66.7±2.4	72.8±2.1	61.8±1.6	69.3±1.4	51.2±2.3	60.8±2.0
CPT-HG	<u>83.8±1.2</u>	83.8±1.0	<u>79.5±1.5</u>	79.3±1.3	72.5±1.9	72.0±1.6	69.0±1.7	68.1±1.4
HeteHG-VAE	82.4±1.1	84.5±0.9	78.2±1.3	80.3±1.1	72.8±1.5	75.0±1.2	68.5±1.4	<u>71.9±1.1</u>
NHGMI	82.6±1.7	83.6±1.4	78.2±1.6	77.9±1.3	71.7±2.1	71.9±1.8	68.1±1.9	67.1±1.6
MEOW	82.3±1.8	82.9±1.5	78.1±2.0	77.5±1.7	71.1±1.3	71.5±1.1	67.5±1.6	66.6±1.3
REHHOR	83.9±1.4*	84.7±1.6*	83.1±1.5*	82.5±1.3*	82.8±1.7*	82.2±1.4*	82.3±1.8*	81.8±1.5*

algorithms. The embedding dimension is kept identical across methods to avoid bias when training downstream classifiers.

For baseline methods, we perform systematic hyperparameter tuning. The search space includes: learning rate $\in \{0.05, 0.01, 0.005, 0.001\}$, negative sample count $\in \{2, 4, 8, 16, 32\}$, weight decay $\in \{0, 10^{-5}, 10^{-4}\}$, and early stopping with patience 20 on validation AUC. For random-walk-based models (node2vec, metapath2vec, HIN2vec, JUST), we search over walk length $\in \{10, 20, 40, 80\}$, walks per node $\in \{5, 10, 20, 40\}$, and window size $\in \{5, 10\}$. For meta-path-based models, we manually enumerate all symmetric meta-paths up to length 4 (e.g., A-P-V-P-A for DBLP) and report the best result. For graph neural networks and contrastive models (R-GCN, HGT, HeteHG-VAE, CPT-HG, NHGMI, MEOW), we search over number of layers $\in \{2, 3\}$, hidden dimension $\in \{64, 128, 256\}$, number of attention heads $\in \{4, 8\}$ (for HGT and CPT-HG), and temperature $\tau \in \{0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$ (for contrastive methods). For each baseline, we run at least 30 hyperparameter trials using random search (or grid search for small spaces) and select the configuration with the highest validation AUC averaged over two initial seeds. The final test results are reported from the best configuration.

All experiments are conducted on a server running Ubuntu 18.04, equipped with an Intel(R) Xeon(R) Silver 4210 CPU (2.20 GHz) and an NVIDIA GeForce RTX 3090 GPU with 24 GB of memory. REHHOR and all deep learning baselines are implemented using the PyTorch framework.

6.4. Node relation prediction

For the general node relation prediction task, we follow the standard experimental protocol established in existing research (Grover & Leskovec, 2016; Zhang & Chen, 2018). Specifically, a varying proportion of edges, ranging from 20% to 80%, is randomly removed from each heterogeneous network to simulate incomplete connectivity. All models are trained on the resulting sparsified networks. During the evaluation phase, the representation of a node pair is computed as the Hadamard product of their respective embeddings. We treat the previously removed edges as positive samples and randomly sample ten times as many non-existent edges from the original network to serve as negative samples. A binary logistic regression classifier is then trained on these relational representations to determine the existence of edges. To ensure statistical reliability, this entire process is repeated ten times, and we report the average performance metrics. The experimental results for the DBLP, PubMed, and Yelp datasets are summarized in Tables 3–5, respectively. Within these tables, the optimal performance in each column is highlighted in bold, while the second-best results are indicated with an underline. * indicates that REHHOR is significantly better than all baselines not in bold.

As evidenced by the performance summarized in Tables 3–5, REHHOR consistently achieves the superior performance across all networks and varying edge deletion ratios. Among the competitive baselines, HeteHG-VAE demonstrates performance closest to REHHOR. This

Table 5
Results of node relation prediction in the yelp network (%).

Deleted Edges	20%		40%		60%		80%	
	AUC	AP	AUC	AP	AUC	AP	AUC	AP
node2vec	51.8±2.5	46.9±2.2	53.0±2.8	47.7±2.4	53.6±2.1	48.3±1.8	52.2±2.6	51.5±2.3
LINE	79.3±1.8	78.9±1.6	77.1±2.2	76.6±1.9	75.0±1.7	74.5±1.5	71.8±2.4	71.5±2.1
TransE	72.3±3.0	68.1±2.7	72.9±2.5	68.7±2.2	73.0±2.0	68.2±1.8	75.5±2.8	71.5±2.5
PTE	50.9±1.5	51.3±1.3	50.7±1.9	51.5±1.6	50.8±1.4	51.3±1.2	50.8±2.1	51.6±1.8
metapath2vec	78.5±2.1	78.4±1.8	74.9±2.4	75.6±2.1	71.3±1.9	71.8±1.6	69.1±2.3	70.6±2.0
HIN2vec	83.9±1.6	84.3±1.4	82.8±1.9	83.4±1.6	82.6±2.2	83.2±1.9	81.0±1.7	81.9±1.5
ConvE	84.0±1.4	84.0±1.2	81.9±1.7	82.1±1.4	79.2±2.0	77.3±1.7	75.9±2.2	76.5±1.9
JUST	49.3±1.2	49.0±1.0	49.3±1.5	49.0±1.3	49.2±1.1	48.9±0.9	49.3±1.8	49.6±1.5
R-GCN	49.8±1.0	49.0±0.8	50.0±1.3	49.1±1.1	49.8±1.6	48.9±1.3	49.6±1.4	48.7±1.2
HGT	81.1±2.3	82.0±2.0	80.2±1.8	81.0±1.6	77.2±2.5	78.0±2.2	74.1±1.9	74.1±1.7
CPT-HG	83.8±1.9	84.4±1.6	78.8±2.2	79.2±1.9	75.4±1.8	75.2±1.5	70.1±2.4	68.7±2.1
HeteHG-VAE	82.4±1.7	83.3±1.4	82.5±2.0	82.6±1.7	82.1±1.5	82.8±1.3	81.6±2.2	82.2±1.9
NHGM	84.1±1.5	84.6±1.3	80.5±1.8	80.6±1.5	76.4±2.1	76.3±1.8	71.8±2.0	71.5±1.7
MEOW	82.2±2.0	83.3±1.7	77.9±2.3	76.7±2.0	74.7±1.6	74.7±1.4	69.6±2.5	68.4±2.2
REHHOR	84.2±1.5*	84.8±1.7*	83.6±1.6*	84.0±1.8*	83.2±1.4*	83.8±1.9*	82.8±2.0*	83.5±1.6*

Table 6
Results of type-specific node relation prediction in the DBLP network (%).

Relation Type	P-A		P-V		P-T	
	AUC	AP	AUC	AP	AUC	AP
node2vec	51.3±3.3	48.7±3.0	66.5±2.8	65.2±2.5	69.3±3.1	69.0±2.7
LINE	66.3±2.7	65.3±2.4	63.5±3.0	61.2±2.5	77.2±2.2	76.0±2.0
TransE	61.2±3.6	61.7±3.1	64.7±3.4	63.3±3.0	75.9±2.8	75.7±2.4
PTE	40.7±1.8	40.3±1.5	41.9±2.1	39.2±1.7	54.6±2.2	52.9±2.0
metapath2vec	62.8±3.9	60.8±3.4	59.9±3.3	59.0±2.8	73.3±3.0	70.9±2.5
HIN2vec	60.5±2.8	60.3±2.5	58.6±3.1	58.3±2.7	70.1±2.4	69.7±2.1
ConvE	59.0±3.0	57.4±2.7	61.4±2.5	57.8±2.2	77.1±2.7	75.7±2.2
JUST	41.9±1.7	41.3±1.4	39.8±2.0	39.6±1.5	52.2±1.8	49.9±1.5
R-GCN	45.9±2.1	45.4±1.8	39.9±2.4	36.8±2.0	53.4±2.5	47.8±2.1
HGT	62.2±3.4	59.8±3.0	53.1±3.7	50.6±3.3	63.2±3.1	65.2±2.7
CPT-HG	70.4±2.2	71.1±2.0	70.6±2.4	65.1±2.1	86.4±1.7	85.4±1.4
HeteHG-VAE	69.1±2.1	67.5±1.8	71.0±2.0	64.3±1.7	85.3±1.8	83.0±1.5
NHGM	68.7±2.4	69.6±2.1	70.7±2.2	64.1±1.8	84.6±2.0	82.6±1.7
MEOW	68.5±2.0	69.1±1.7	70.3±2.1	63.8±1.8	83.7±2.1	81.8±1.8
REHHOR	70.9±1.5*	71.7±1.6*	71.2±1.3*	65.5±1.7*	87.1±1.1*	86.9±1.2*

relative success is primarily attributed to its strategy of mapping heterogeneous networks into hypergraphs, which enables the capture of higher-order structural information. In contrast, while CPT-HG performs well under low edge deletion ratios, its accuracy significantly deteriorates as the deletion ratio increases. This decline likely stems from its subgraph-based pre-training tasks, which rely heavily on the availability of a relatively complete network topology.

Notably, certain heterogeneous representation learning methods, such as R-GCN, exhibit lower predictive accuracy compared to homogeneous counterparts. This observation suggests that focusing exclusively on specific semantics, such as relation types or predefined patterns, may be less effective for node-level proximity tasks than the direct capture of local neighborhood structures. Furthermore, on the Yelp dataset, methods like HIN2vec and ConvE significantly outperform their respective results on DBLP and PubMed. This discrepancy is largely rooted in the highly imbalanced distribution of relation types within the Yelp network, coupled with the pair-reconstruction strategies of these models. Since business-review interactions constitute the vast majority of the network, these methods predominantly prioritize the reconstruction of this specific category during training, thereby yielding inflated performance metrics on that domain.

Compared with other competitive baselines, REHHOR exhibits significantly smaller performance fluctuations across various edge deletion ratios. This stability stems from the structural properties of heterogeneous networks, where extensive edge removal (e.g., an 80% deletion ratio) severely degrades local topological information while leaving the network backbone-formed by HOU and their interactions-relatively in-

tact. Consequently, preserving such backbone information is essential for ensuring that representation learning methods can effectively support relation prediction in sparse environments.

Through hierarchical encoding on Node-HOU bipartite graphs and the explicit reconstruction of HOU interactions, REHHOR concurrently captures both the global backbone structure and the local node-level associations. In contrast, while HeteHG-VAE utilizes hyperedges, it focuses on directly reconstructing the relations between nodes and hyperedges, which primarily encapsulates local interactions similar to the node-HOU level. As the edge deletion ratio increases, the rapid loss of local connectivity leads to a pronounced decline in the effectiveness of HeteHG-VAE.

In summary, the superior performance and robustness of REHHOR arise from its faithful simulation of the hierarchical generation process of heterogeneous networks. This approach establishes a cohesive representational link from macro-level HOU interactions to micro-level node-HOU relations, thereby maintaining structural integrity even under significant data scarcity.

6.5. Type-specific node relation prediction

In practical applications, the relations between nodes are typically constrained by their semantic types. To evaluate model performance under these specialized conditions, this section conducts experiments focusing on node relations of specific types. Following the established protocol, we randomly remove 20% of the edges for a designated relation type within each heterogeneous network. The models are then trained and evaluated using the same procedure as described for general node

Table 7
Results of type-specific node relation prediction in the PubMed network (%).

Relation Type	G-C		G-D		G-S	
	AUC	AP	AUC	AP	AUC	AP
node2vec	70.6±2.3	68.1±2.0	73.8±2.1	76.2±1.8	75.0±1.9	72.1±1.6
LINE	73.2±1.9	77.1±1.6	75.0±2.0	78.1±1.7	79.8±1.5	80.8±1.3
TransE	69.4±2.5	71.7±2.2	72.8±2.3	73.9±2.0	72.8±2.1	75.0±1.8
PTE	56.2±1.4	57.6±1.2	55.1±1.6	54.9±1.3	56.7±1.5	57.7±1.2
metapath2vec	76.5±2.2	77.8±1.9	72.6±2.4	79.1±2.1	77.3±2.0	79.3±1.7
HIN2vec	51.7±2.0	53.6±1.7	53.5±2.2	55.0±1.9	49.6±2.1	54.6±1.8
ConvE	71.8±1.8	76.0±1.5	75.1±1.7	77.3±1.4	74.0±1.9	76.4±1.6
JUST	56.7±1.1	50.8±0.9	48.8±1.3	50.7±1.0	50.7±1.2	52.5±1.0
R-GCN	42.8±1.5	45.9±1.2	44.8±1.6	47.8±1.3	45.8±1.4	48.5±1.1
HGT	73.7±2.4	77.0±2.1	78.5±2.0	76.9±1.7	68.5±2.3	75.7±2.0
CPT-HG	85.5±1.3	86.3±1.1	83.6±1.4	87.3±1.2	84.1±1.2	84.2±1.0
HeteHG-VAE	83.6±1.2	84.8±1.0	84.7±1.1	84.7±0.9	83.4±1.3	85.7±1.1
NHGM	84.9±1.5	85.4±1.3	82.5±1.6	84.9±1.4	82.5±1.5	82.8±1.2
MEOW	84.1±1.4	84.5±1.2	81.9±1.5	83.5±1.3	82.8±1.4	82.5±1.1
REHHOR	85.7±1.4*	86.5±1.6*	85.8±1.5*	87.2±1.3*	84.0±1.7*	85.5±1.4*

Table 8
Results of type-specific node relation prediction in the yelp network (%).

Relation Type	B-L		B-P		B-S	
	AUC	AP	AUC	AP	AUC	AP
node2vec	74.8±2.4	78.4±2.1	76.9±2.2	76.6±1.9	66.9±2.0	66.1±1.7
LINE	80.6±1.8	83.1±1.6	79.1±2.0	<u>83.9±1.7</u>	72.1±1.9	73.8±1.6
TransE	81.9±2.2	83.0±1.9	79.8±2.1	80.6±1.8	67.4±2.3	68.8±2.0
PTE	67.3±1.6	64.5±1.4	64.4±1.8	62.8±1.5	49.3±1.3	51.2±1.1
metapath2vec	85.9±1.7	84.3±1.5	82.1±1.9	81.1±1.6	72.3±2.0	71.0±1.7
HIN2vec	58.2±2.0	59.4±1.7	66.6±2.2	54.5±1.9	57.5±2.1	47.2±1.8
ConvE	84.0±1.6	82.0±1.4	80.1±1.8	78.7±1.5	71.8±1.9	69.0±1.6
JUST	54.6±1.2	55.1±1.0	52.7±1.4	49.7±1.1	49.8±1.2	47.8±1.0
R-GCN	50.4±1.4	51.8±1.2	46.7±1.6	48.9±1.3	44.4±1.5	43.0±1.2
HGT	74.8±2.5	77.7±2.2	68.8±2.6	71.1±2.3	70.7±2.4	67.3±2.1
CPT-HG	92.3±1.2	90.8±1.0	84.1±1.3	82.5±1.1	<u>77.6±1.4</u>	72.6±1.2
HeteHG-VAE	91.4±1.3	88.4±1.1	83.1±1.4	83.4±1.2	75.0±1.5	71.3±1.3
NHGM	<u>92.7±1.1</u>	<u>91.3±0.9</u>	83.8±1.2	82.5±1.0	77.3±1.3	72.4±1.1
MEOW	91.7±1.4	90.8±1.2	83.1±1.5	83.0±1.3	76.1±1.6	71.7±1.4
REHHOR	93.3±1.5*	91.9±1.7*	<u>84.0±1.6*</u>	84.9±1.4*	77.9±1.8*	<u>72.5±1.5*</u>

Table 9
Results of HOU interaction relation prediction in the DBLP (%).

Relation Type	ξ -A- ξ		ξ -V- ξ		ξ -T- ξ	
	AUC	AP	AUC	AP	AUC	AP
node2vec	50.6±3.5	51.2±3.2	57.6±2.8	56.0±2.5	52.2±3.0	51.8±2.7
LINE	66.6±2.4	66.7±2.1	59.9±2.9	58.8±2.6	59.5±2.5	61.4±2.2
TransE	63.0±3.0	63.1±2.7	62.4±3.2	64.7±2.8	66.9±2.6	67.6±2.3
PTE	41.8±2.0	43.8±1.7	46.6±2.3	46.5±2.0	53.2±1.8	51.4±1.5
metapath2vec	65.9±3.2	67.2±2.9	64.4±3.0	63.7±2.7	68.8±2.5	68.5±2.2
HIN2vec	58.5±2.6	58.5±2.3	56.0±2.9	55.4±2.5	59.5±2.4	59.3±2.1
ConvE	61.5±2.5	61.2±2.2	62.9±2.7	61.5±2.4	67.9±2.3	65.8±2.0
JUST	43.4±1.6	43.5±1.3	42.8±1.9	40.8±1.6	49.4±1.7	49.3±1.4
R-GCN	46.6±1.9	47.6±1.6	44.4±2.2	46.3±1.8	51.4±2.0	50.6±1.7
HGT	73.4±2.2	72.5±2.0	67.4±2.5	66.7±2.2	68.9±2.4	69.4±2.1
CPT-HG	75.8±2.0	72.4±1.8	76.7±2.1	76.8±1.9	72.6±2.2	69.9±1.9
HeteHG-VAE	<u>76.7±1.7</u>	<u>76.1±1.5</u>	<u>78.8±1.8</u>	<u>79.3±1.6</u>	<u>80.5±1.4</u>	<u>80.3±1.2</u>
NHGM	74.7±2.1	71.5±1.9	76.1±2.0	76.3±1.7	72.4±2.0	69.4±1.7
MEOW	73.4±2.3	71.7±2.0	74.7±2.2	75.1±1.9	72.1±2.1	68.9±1.8
REHHOR	81.8±1.4*	81.8±1.5*	83.2±1.2*	83.1±1.3*	87.9±0.9*	87.6±1.0*

relation prediction, with the average results across ten independent runs being reported. The prediction performance on the DBLP, PubMed, and Yelp networks is summarized in Tables 6–8, respectively. In these tables, the optimal result in each category is highlighted in bold, while the second-best performance is indicated with an underline. * indicates that REHHOR is significantly better than all baselines not in bold.

Experimental results indicate that REHHOR consistently achieves superior performance across nearly all semantic relations. CPT-HG ex-

hibits competitive accuracy in these tasks, which is largely attributable to its collaborative optimization of edge-level and subgraph-level pre-training. However, the performance of CPT-HG tends to fluctuate significantly as the network becomes sparser, whereas HeteHG-VAE remains a more stable baseline across all datasets. Despite this stability, HeteHG-VAE still fails to match the performance of REHHOR.

The advantage of REHHOR in this tasks stems from the structural insights gained through Node-HOU bipartite graphs. This architecture

Table 10
Results of HOU interaction relation prediction in the PubMed network (%).

Relation Type	ξ -C- ξ		ξ -D- ξ		ξ -S- ξ	
	AUC	AP	AUC	AP	AUC	AP
node2vec	66.8±2.8	67.6±2.5	65.0±2.6	65.1±2.3	63.2±2.7	62.7±2.4
LINE	66.6±2.5	67.1±2.2	64.1±2.7	65.3±2.4	63.4±2.6	61.1±2.3
TransE	64.5±2.9	64.4±2.6	62.2±3.0	62.3±2.7	67.6±2.5	66.5±2.2
PTE	47.3±1.8	47.4±1.6	49.0±2.0	50.0±1.7	46.6±1.9	48.2±1.6
metapath2vec	<u>77.7±2.0</u>	<u>79.1±1.7</u>	72.7±2.3	71.5±2.0	78.6±1.9	79.3±1.6
HIN2vec	58.9±2.4	59.2±2.1	57.3±2.6	57.5±2.3	62.7±2.2	63.6±1.9
ConvE	66.8±2.3	66.5±2.0	62.5±2.5	61.0±2.2	65.1±2.4	63.2±2.1
JUST	44.8±1.5	44.7±1.3	43.2±1.7	43.8±1.4	48.3±1.6	49.2±1.3
R-GCN	56.3±1.9	56.1±1.6	53.8±2.1	55.9±1.8	52.2±2.0	51.0±1.7
HGT	71.3±2.4	72.7±2.1	68.1±2.6	65.7±2.3	74.4±2.2	73.6±1.9
CPT-HG	69.0±2.2	69.3±2.0	68.1±2.4	68.9±2.1	72.9±2.1	71.7±1.8
HeteHG-VAE	76.7±1.8	76.8±1.6	<u>78.8±1.7</u>	<u>80.1±1.4</u>	<u>80.5±1.5</u>	<u>81.2±1.2</u>
NHGMI	68.0±2.3	67.7±2.0	67.2±2.5	66.7±2.2	72.3±2.1	71.2±1.8
MEOW	67.2±2.2	67.0±1.9	66.5±2.4	65.8±2.1	71.2±2.0	70.7±1.7
REHHOR	85.3±1.4*	84.8±1.6*	83.2±1.5*	82.5±1.3*	87.8±1.2*	86.9±1.4*

Table 11
Results of HOU interaction relation prediction in the yelp network (%).

Relation Type	ξ -L- ξ		ξ -P- ξ		ξ -S- ξ	
	AUC	AP	AUC	AP	AUC	AP
node2vec	52.4±2.5	52.1±2.3	55.2±2.6	54.7±2.4	57.1±2.4	54.6±2.2
LINE	64.5±2.0	61.6±1.8	56.8±2.2	56.9±2.0	61.2±2.1	60.4±1.9
TransE	63.5±2.3	62.9±2.0	61.0±2.4	60.0±2.1	60.4±2.3	60.0±2.0
PTE	49.3±1.6	48.8±1.4	49.0±1.8	47.0±1.5	50.3±1.7	48.5±1.4
metapath2vec	71.9±2.1	71.4±1.9	68.8±2.2	68.2±2.0	71.0±2.0	69.9±1.8
HIN2vec	72.8±1.9	72.2±1.7	70.9±2.0	70.1±1.8	73.0±1.9	71.3±1.7
ConvE	72.6±2.0	71.3±1.8	74.1±1.9	71.2±1.7	70.4±2.1	69.9±1.8
JUST	49.0±1.4	45.6±1.2	47.2±1.6	47.0±1.3	49.2±1.5	46.3±1.2
R-GCN	52.2±1.7	52.7±1.5	49.2±1.9	51.8±1.6	50.1±1.8	50.6±1.5
HGT	72.0±2.2	71.5±2.0	73.9±2.1	73.2±1.8	73.5±2.0	71.9±1.7
CPT-HG	71.0±2.1	70.3±1.9	71.5±2.2	71.3±2.0	69.5±2.1	68.1±1.8
HeteHG-VAE	78.6±1.6	76.7±1.4	<u>79.7±1.5</u>	<u>78.7±1.3</u>	<u>76.6±1.7</u>	<u>75.7±1.5</u>
NHGMI	79.0±1.5	77.5±1.3	79.4±1.6	77.0±1.4	75.5±1.6	74.1±1.4
MEOW	73.9±1.8	72.8±1.6	76.0±1.7	75.0±1.5	75.7±1.7	74.3±1.5
REHHOR	82.8±1.5*	81.8±1.7*	84.9±1.4*	84.1±1.6*	83.1±1.5*	82.8±1.4*

allows the model to decouple and capture the distinct interaction patterns associated with different node types, thereby constructing a multifaceted network backbone. By integrating these global skeletal features with local connectivity, REHHOR more effectively constrains the representation learning process within specific semantic contexts, leading to more accurate link inference than methods relying on simpler local or hypergraph reconstructions.

6.6. HOU interaction relation prediction

To verify whether REHHOR and the baselines can effectively characterize the hierarchical generation process of heterogeneous networks, we design an interaction relation prediction experiment at the HOU level. Specifically, for a given inter-node relation type ($o_c - o$), we randomly remove 20% of the corresponding HOU interaction relations ($\xi - o - \xi$). In the DBLP network, for instance, this involves deleting three distinct interaction types: collaborative writing (ξ -A- ξ), co-publication (ξ -V- ξ), and shared topics (ξ -T- ξ).

Models are applied to the processed networks to generate representation vectors. For REHHOR, we directly utilize the HOU representations obtained via the pooling operation in Eq. (10). For the baseline methods, HOU representations are constructed by concatenating the center node embedding with the sum of its associated non-central node embeddings, mimicking the REHHOR architecture for a fair comparison. The interaction prediction is then performed using the Hadamard product of HOU pairs followed by a logistic regression classifier.

Average results across ten independent trials for the DBLP, PubMed, and Yelp networks are summarized in Tables 9–11, respectively. In these tables, the optimal performance is highlighted in bold, while the second-best results are underlined. * indicates that REHHOR is significantly better than all baselines not in bold. REHHOR achieves substantial performance gains over all baselines across all datasets, which demonstrates its superior ability to characterize the network's hierarchical generation process. Notably, HeteHG-VAE also outperforms other baselines, benefiting from its hypergraph-based approach that captures higher-order dependencies to a certain extent.

6.7. 2D visualization of HOU embeddings

To qualitatively evaluate the effectiveness of the skeletal reconstruction, we project the learned HOU embeddings onto a two-dimensional space using t-SNE. Fig. 5(a)–(c) show the results on DBLP for HOU embeddings derived from Author (A), Venue (V), and Term (T) interactions, respectively. Fig. 5(d)–(f) show the corresponding results on Yelp for Location (L), Review (P), and Star (S). Clear clustering patterns are observed across all types, demonstrating that HOU embeddings with similar functional roles are grouped together. In DBLP, the Author-based embeddings appear dispersed but consist of tightly overlapping points, indicating precise semantic alignment. These visualizations demonstrate that the hierarchical skeletal learning effectively captures both structural and semantic similarities among higher-order units.

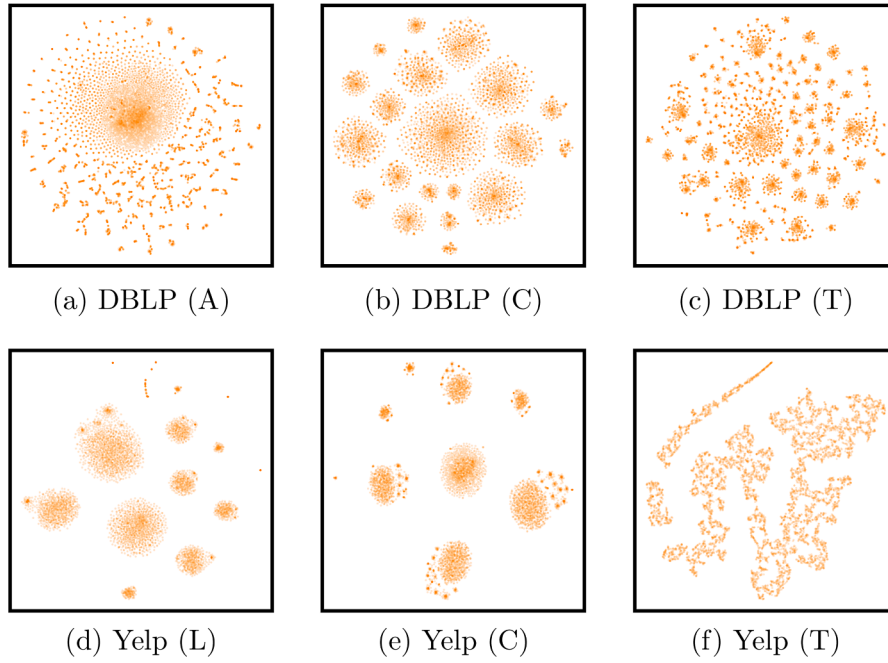


Fig. 5. 2D visualization of HOU embeddings based on T-SNE algorithm.

Table 12
Node relation prediction on the hetionet network (%).

Method	AUC	AP
HGT	60.1 $\pm$ 2.8	58.4 $\pm$ 2.6
CPT-HG	64.3 $\pm$ 2.5	62.7 $\pm$ 2.3
HeteHG-VAE	67.5 $\pm$ 2.2	66.1 $\pm$ 2.0
REHHOR	71.2 $\pm$ 2.0*	69.8 $\pm$ 2.1*

6.8. Prediction on multi-center network

To evaluate the generalizability of REHHOR to multi-center heterogeneous networks, we construct a multi-center subgraph from Hetionet. We compare REHHOR against three representative deep learning baselines, including HGT, CPT-HG, and HeteHG-VAE. All models are trained on the same training split with 20% of edges randomly removed for each relation type, except for the G-C (gene-compound) edges, which are kept intact to preserve the connectivity of the center node set. For HOU interaction relation prediction, we evaluate four types of HOU interactions corresponding to the non-center node types: disease (ξ -D- ξ), side effect (ξ -S- ξ), pathway (ξ -P- ξ), and anatomy (ξ -A- ξ). We report the average results over five independent runs with standard deviations. The data annotation procedure follows the same protocol as in previous subsections.

Table 12 reports the node relation prediction results. REHHOR outperforms all baselines. Table 13 presents the HOU interaction relation prediction results across four non-center types. Two main observations can be made. First, REHHOR consistently outperforms all baselines on every HOU interaction type. Second, the performance gap between REHHOR and baselines widens as the non-center node set becomes smaller, suggesting that REHHOR is particularly effective at capturing fine-grained skeletal structures in sparse regimes. These results collectively demonstrate that REHHOR's hierarchical generative framework extends naturally to multi-center heterogeneous networks and retains its advantages.

6.9. Ablation experiment

To investigate the individual contributions of the core components in REHHOR, we design the following variant models for comparative analysis. Each variant removes or replaces a specific design element while keeping all other settings unchanged. Subsequently, we conduct general node relation prediction experiments comparing REHHOR with these variants on the DBLP and Yelp networks across different edge deletion ratios. The results are illustrated in Fig. 6.

- **REHHOR w/o HOU.** Removes HOU extraction entirely and models relations directly at the node-pair level, using a conventional node relation reconstruction loss.
- **REHHOR w/o Type-Specific.** Merges all node-HOU bipartite graphs into a single bipartite graph (i.e., a single $\mathbf{B}$ matrix), ignoring type distinctions among non-center nodes.
- **REHHOR w/o Directed Noise.** Replaces the data-dependent directional noise with isotropic Gaussian noise, i.e., standard diffusion noise.
- **REHHOR w/o DM.** Replaces the directed diffusion-based encoder with a standard MLP, evaluating the impact of the diffusion process on node representation quality.
- **REHHOR w/o Residual.** Removes the residual connections from the MLP denoising encoder.
- **REHHOR w/ Sum Pooling.** Replaces the pooling with sum pooling over all nodes in the HOU.
- **REHHOR w/o HOC.** Substitutes the HOU interaction reconstruction module with a traditional inter-node relation reconstruction objective, testing the necessity of modeling the high-order network backbone.

The experimental results across both networks reveal clear differences in the contribution of each component to REHHOR's performance. The full REHHOR model achieves the best results across all edge deletion ratios, confirming the synergistic effect of all design elements.

Removing residual connections (w/o Residual) causes a moderate performance drop, indicating that residual connections help stabilize training and preserve gradient flow in the denoising MLP, but their absence does not catastrophically harm the model. Replacing the center-anchored concatenation pooling with simple sum pooling (w/ Sum

Table 13
HOU interaction relation prediction on the hetionet network (%).

Method	ξ -D- ξ (D: 136)		ξ -S- ξ (S: 5,701)		ξ -P- ξ (P: 1,822)		ξ -A- ξ (A: 400)	
	AUC	AP	AUC	AP	AUC	AP	AUC	AP
HGT	68.2 $\pm$ 3.0	66.5 $\pm$ 2.8	73.8 $\pm$ 2.6	72.1 $\pm$ 2.4	70.5 $\pm$ 2.9	68.9 $\pm$ 2.7	72.0 $\pm$ 2.5	70.4 $\pm$ 2.3
CPT-HG	72.6 $\pm$ 2.6	71.0 $\pm$ 2.4	76.4 $\pm$ 2.3	74.8 $\pm$ 2.1	74.0 $\pm$ 2.5	72.3 $\pm$ 2.3	75.2 $\pm$ 2.2	73.6 $\pm$ 2.0
HeteHG-VAE	76.3 $\pm$ 2.3	74.9 $\pm$ 2.1	79.5 $\pm$ 2.0	78.2 $\pm$ 1.8	77.8 $\pm$ 2.2	76.4 $\pm$ 2.0	78.9 $\pm$ 2.0	77.5 $\pm$ 1.8
REHHOR	84.1 $\pm$ 2.1*	82.7 $\pm$ 2.2*	81.2 $\pm$ 2.4*	79.9 $\pm$ 2.2*	83.5 $\pm$ 2.3*	82.1 $\pm$ 2.4*	83.8 $\pm$ 2.1*	82.4 $\pm$ 2.2*

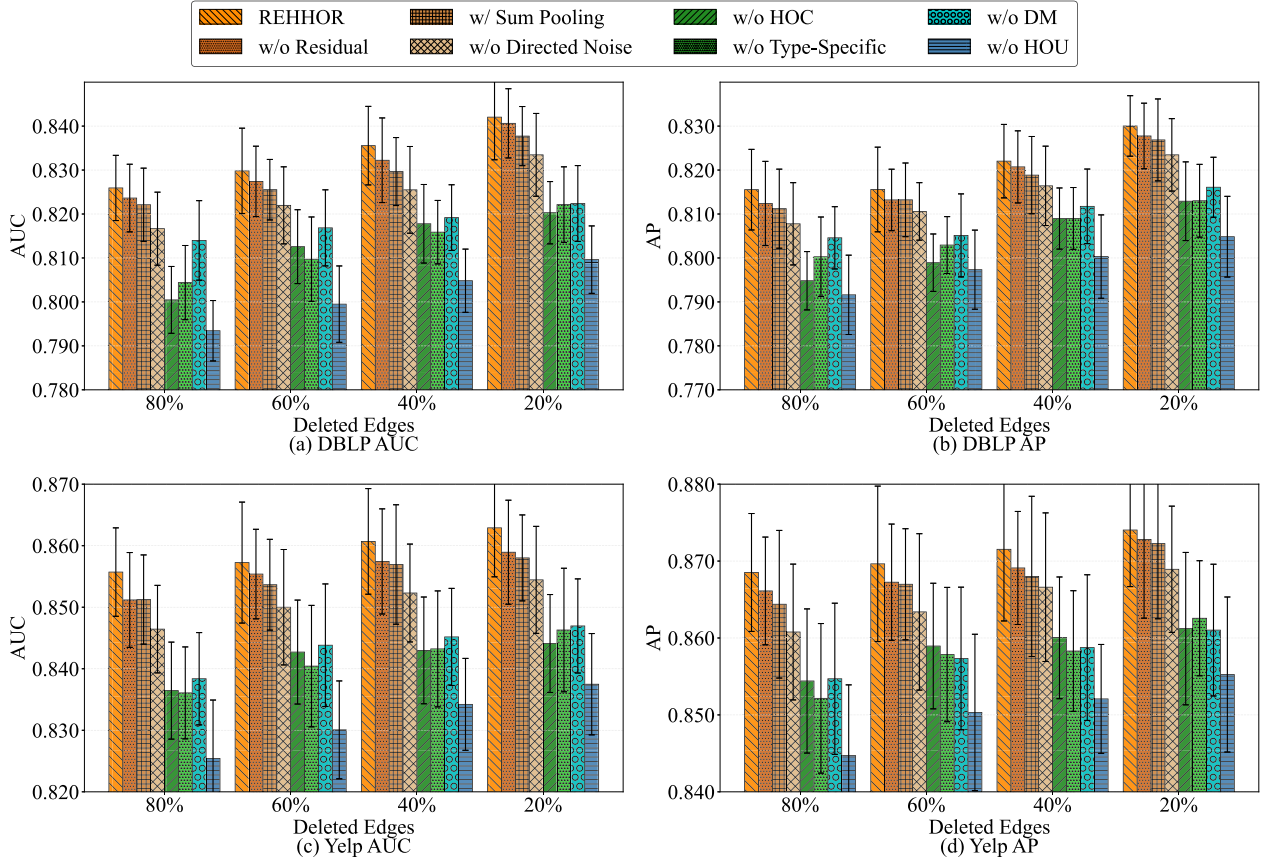


Fig. 6. Results of ablation experiments.

Pooling) leads to a slightly larger decline, suggesting that explicitly preserving the center node identity is beneficial for capturing the functional role of HOU.

A more significant performance degradation occurs when the directed noise is replaced by isotropic Gaussian noise (w/o Directed Noise). This result supports our claim that data-dependent anisotropic noise is crucial for preserving weak structural signals, especially under high sparsity. Notably, both the removal of the HOU interaction contrastive loss (w/o HOC) and the removal of type-specific bipartite decomposition (w/o Type-Specific) yield comparable performance reductions. This indicates that skeletal-level optimization and type-aware decomposition are both essential for capturing heterogeneous semantics, and their contributions are of similar magnitude.

The two most severe performance drops are observed when the diffusion mechanism is completely removed (w/o DM) and when the entire HOU hierarchy is discarded (w/o HOU). The fact that w/o DM (using a plain MLP encoder) still outperforms most baseline methods confirms that the HOU interaction reconstruction module alone provides substantial benefits by capturing the network backbone. However, the worst performance of w/o HOU validates that the hierarchical HOU structure is the most critical component of REHHOR. Without HOU, the model

reverts to flat node-pair relation reconstruction, losing the ability to capture higher-order functional units and their interactions. This finding proves that the hierarchical HOU structure is not a mere inductive bias but a necessary condition for capturing the network's generative process.

Furthermore, the performance gain of REHHOR over w/o DM is more pronounced on the DBLP dataset than on the Yelp dataset. This discrepancy is likely due to the relatively uniform distribution of relation types in Yelp, where the diffusion-based encoder has less opportunity to leverage its superior encoding capacity compared to the more complex semantic structure of DBLP. It is also important to observe that even without the diffusion module, w/o DM maintains a performance edge over most baseline methods. This success is directly attributable to the HOU interaction reconstruction module, which ensures that the model successfully captures the underlying backbone structure even with a simpler encoding architecture.

6.10. Analysis on central node type selection

To evaluate how sensitive REHHOR is to the choice of central node type, we conduct experiments on the DBLP dataset where we artificially

Table 14
Results of different choices of center nodes on the DBLP network.

Center Type	20% deletion		80% deletion	
	AUC	AP	AUC	AP
Paper (P, default)	0.843 ± 0.009	0.830 ± 0.011	0.826 ± 0.008	0.815 ± 0.010
Author (A)	0.752 ± 0.018	0.731 ± 0.020	0.698 ± 0.022	0.684 ± 0.019
Venue (V)	0.691 ± 0.016	0.688 ± 0.017	0.672 ± 0.021	0.655 ± 0.018
Term (T)	0.735 ± 0.019	0.719 ± 0.018	0.681 ± 0.020	0.662 ± 0.022

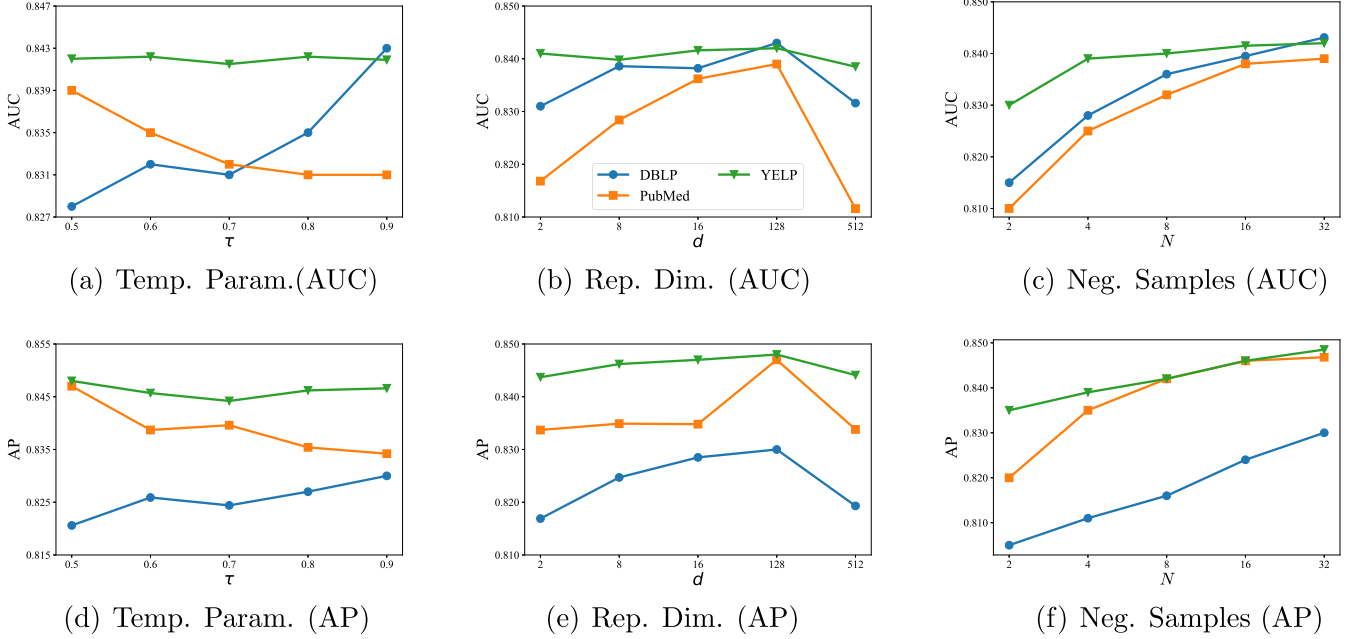


Fig. 7. Sensitivity results on hyperparameters.

varied the designated center type. DBLP naturally has Paper (P) as the unique center. We compare three alternative settings: (i) using Author (A) as center, (ii) using Venue (V) as center, and (iii) using Term (T) as center. For each setting, we extract HOU according to the alternative center whose embedding is the mean of corresponding type of node embeddings. Node relation prediction is performed under 20% and 80% edge deletion ratios. The results are shown in Table 14.

The results show that using the correct center type (Paper) yields the best performance. Using alternative center types leads to a noticeable drop, indicating that REHHOR is moderately sensitive to center selection. The drop is more pronounced under high sparsity (80% edge deletion), as the wrong center type produces less coherent HOU that fail to capture the true generative structure.

6.11. Parameter sensitivity analysis

This section investigates the impact of the contrastive loss temperature parameter τ and the representation dimension d on the performance of REHHOR. We conduct general node relation prediction on all three datasets with a 20% edge deletion ratio. In each experiment, one parameter is varied while the other remains fixed at its default value. The observed performance trends are illustrated in Fig. 7 and analyzed below.

Impact of Temperature Parameter τ . The sensitivity of REHHOR to τ varies across datasets due to their distinct relation distributions. According to Eq. (13), a smaller τ value sharpens the similarity scores between HOU representations. On the DBLP network, where P-T relations dominate, a larger τ helps alleviate the excessive aggregation of representations caused by this imbalance, thereby improving results. Conversely, in the PubMed network, where G-S relations are relatively sparse compared to other types, a smaller τ proves beneficial for cap-

turing the network backbone more effectively. On the Yelp network, the overwhelming monopoly of B-P relations renders the model relatively insensitive to changes in τ . Consequently, τ is strategically tuned for each network to accommodate these specific structural characteristics.

Impact of Representation Dimension d . As d increases, the prediction accuracy across all networks follows a consistent trend of rising and then declining, typically peaking at $d = 128$. This suggests that while a sufficiently high dimension is necessary to encode complex structural information, an excessively large d leads to overfitting. The Yelp network exhibits lower sensitivity to changes in d than DBLP or PubMed, a phenomenon further attributed to its highly skewed relation distribution which limits the semantic diversity required for higher-dimensional representations.

Impact of Negative Sampling Number N . The number of negative samples N plays a critical role in the stability and convergence of the contrastive learning objective. As illustrated in Fig. 7, the model performance generally improves as N increases from a small value, eventually reaching a plateau. With a limited N (e.g., $N < 8$), the model struggles to approximate the full softmax distribution, leading to suboptimal discrimination capabilities. Increasing N provides a richer set of contrasting signals, enabling REHHOR to push apart non-interacting node pairs more effectively and learn a more uniform representation distribution. However, beyond a certain threshold (typically $N \geq 16$), the marginal gain diminishes. This indicates that the model has captured sufficient discriminative information, and further increasing N yields redundant gradients while incurring higher computational costs.

6.12. Computational efficiency analysis

To evaluate the computational efficiency of the proposed method, we compare the training time per epoch of REHHOR against three deep

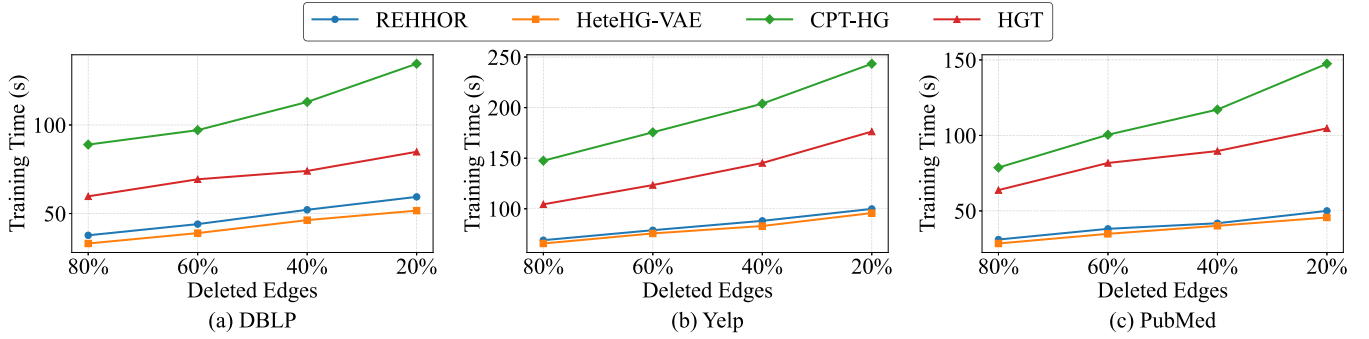


Fig. 8. Training time comparison.

Table 15

Parameter number comparison on the DBLP network.

Method	Number of Parameters (Millions)
REHHOR	46
HeteHG-VAE	39
CPT-HG	147
HGT	134

learning baseline models, including HeteHG-VAE, CPT-HG, and HGT. The comparison is conducted across DBLP, Yelp, and PubMed datasets and under varying network densities by randomly removing edges.

As shown in Fig. 8, the results indicate that REHHOR consistently demonstrates superior computational efficiency compared to graph neural network models CPT-HG and HGT. It achieves training speeds comparable to those of HeteHG-VAE. As the network density increases, the training time of all models grows. However, the increase for REHHOR remains gentle, with a relatively small slope. Our directed diffusion mechanism fundamentally differs from standard diffusion models. Instead of performing iterative noise prediction over dozens of steps, REHHOR directly reconstruct HOU interactions via a contrastive objective. Consequently, both the time and space overhead of our method are significantly lower than those of generic diffusion models, and its complexity is comparable to that of VAE-based approaches such as HeteHG-VAE. In contrast, graph neural network models incur the highest computational cost, due to the complexity of message passing.

We also compare the number of trainable parameters of REHHOR with the baseline models on the DBLP dataset, as shown in Table 15. The results demonstrate that REHHOR maintains a compact model architecture with approximately 46 million parameters. This is significantly lower than the complex graph neural network baselines. Although REHHOR has slightly more parameters than HeteHG-VAE, it achieves a better balance between model capacity and complexity. This indicates that REHHOR can effectively capture higher-order structural information without incurring the excessive parameter overhead.

7. Conclusion and future work

In this paper, we have presented REHHOR, a novel framework for heterogeneous network representation learning that transcends the limitations of traditional pairwise modeling by adopting a hierarchical generative perspective. By integrating type-specific bipartite decomposition with a directed diffusion mechanism and a skeletal reconstruction objective, REHHOR effectively preserves semantic diversity, addresses anisotropic structural patterns, and ensures global topological integrity even under data sparsity. Extensive evaluations on four real-world datasets, including a multi-center Hetionet subset, confirm that this hierarchical approach significantly enhances robustness, consistently achieving state-of-the-art performance across node relation pre-

diction, type-specific relation prediction, and HOU interaction prediction tasks.

Advantages in Relation to Similar Schemes. Compared with existing methods, REHHOR offers several distinct advantages. First, unlike meta-path-based methods such as metapath2vec and HIN2vec that flatten higher-order interactions into sequences, REHHOR explicitly models the hierarchical generation process and preserves the global skeleton even when eighty percent of edges are deleted. Second, in contrast to hypergraph-based approaches like HeteHG-VAE that only reconstruct node-hyperedge associations, REHHOR optimizes a skeletal contrastive objective that captures inter-HOU interactions, leading to more robust representations. Third, while contrastive methods such as CPT-HG rely on subgraph sampling that degrades under edge deletion, REHHOR's directed diffusion mechanism injects data-dependent anisotropic noise, maintaining structural signals when local connectivity is weak. Fourth, the type-specific bipartite decomposition avoids the semantic homogenization that occurs when collapsing heterogeneous nodes into a unified hyperedge. As shown in our ablation study, removing any of these components causes a significant performance drop, confirming their individual necessity.

Limitations and Future Directions. Despite its strengths, REHHOR has several limitations. First, it assumes the network contains central node types or a minimal center set that anchor higher-order units. For networks that genuinely lack any recurring functional pattern, a scenario that is rare in practice, the framework falls back to an edge-level model treating each node as its own HOU, thereby losing the hierarchical benefits. Second, the directed noise statistics μ and σ are computed from the training network; consequently, if the training distribution is not representative of the test distribution due to extreme domain shift, performance may degrade. Third, the current framework is designed for static networks and does not naturally capture temporal dynamics.

These limitations point to promising future research directions. We plan to extend REHHOR to dynamic heterogeneous networks by modeling the evolution of HOUs over time, for instance through temporal point processes or recurrent diffusion mechanisms. Additionally, we aim to explore adaptive HOU discovery that does not rely on predefined central node types, using graph neural networks or clustering algorithms to identify functional units end-to-end. Another direction is to integrate REHHOR with large-scale graph foundation models to achieve more generalized and transferable structural inference across diverse network domains.

CRediT authorship contribution statement

Xuan Guo: Conceptualization, Methodology, Writing – original draft; **Qiyao Peng:** Investigation, Methodology; **Wenjun Wang:** Funding acquisition; **Yaozhi Zhang:** Visualization, Validation; **Zihao Liang:** Writing – review & editing; **Wei Yu:** Writing – review & editing; **Ying Zhao:** Funding acquisition; **Tianpeng Li:** Project administration.

Data availability

The data that has been used is confidential.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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